

On the variable strength of *only**

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A distributed analysis of *only* divides its contribution between two expressions, both of which quantify over alternatives: a subtraction operator and an exhaustification operator. This dual quantification introduces two points at which a restriction of the domain of alternatives can affect the semantic contribution of *only* to the sentence. We provide a new argument in support of such double dependence on alternatives, drawing on the well-known variability in the strength of the positive presupposition associated with *only* – in some cases the prejacent appears to be presupposed, while in others the presupposition is weaker, requiring only that either the prejacent or some other focus alternative be true (e.g., Horn 1969 vs. Horn 1996).

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1 The background

Only gives rise to two kinds of inferences: a positive inference and an exhaustive inference. In (1), the positive inference is that Gal was at the party – that is, the prejacent of *only*. The exhaustive inference is that no one distinct from Gal was at the party. The positive inference has received considerable attention in the literature owing to its complex and challenging nature (e.g., Horn 1969, 1972, 1996, Atlas 1993, Klinedinst 2005, Roberts 2006, 2011, Wagner 2006, von Stechow & Iatridou 2007, van Rooij & Schulz 2007, Ippolito 2008, Beaver & Clark 2008, among others).

(1) Only Gal_F was at the party.

Positive inference: Gal was at the party

Exhaustive inference: No one distinct from Gal was at the party

*Acknowledgments to be added. The paper is a slight revision of Sects. 1.2, 5, and 6.2 of Crnić 2022. There are some additions: we discuss a criticism of this earlier paper (Alonso-Ovalle & Hirsch 2023), apply some new tests that have been developed in the meantime (Doron & Wehbe 2022, Guerrini & Wehbe 2025), point to new insights pertaining to sufficiency modal constructions (Condoravdi & Francez 2022), and discuss one integrated analysis of *only* in more depth (Beaver & Clark 2008). There are also some reductions: the mechanisms employed in the earlier and current derivations (MIN in Sect. 5 and *pex* in Sect. 6.2 of Crnić 2022) are not discussed in any depth or at all, hopefully without compromising the force of the paper (see also Crnić 2024 for a discussion of them).

The analyses of *only* can be broadly split into two camps: integrated vs. distributed analyses.

1.1 An integrated analysis of *only*

The benchmark analysis of *only* is due to Horn 1969. He takes both inferences in (1) to be generated by *only*, which is why we label the analysis as ‘integrated’. The definition of *only* is given in (2), where *only* is treated as a sentential operator (see, e.g., Rooth 1992, Hirsch 2017). The positive inference is presupposed – namely, that the prejacent of *only* is true. The exhaustive inference is asserted – namely, that no focus alternative to the sister of *only* is true if it is relevant (that is, if it is in the resource domain of *only*, C) and excludable (that is, if it belongs to every maximal set of alternatives that can be jointly negated while keeping the assertion true; see, e.g., Fox & Hackl 2006, Fox 2007, Alonso-Ovalle 2008, Chierchia et al. 2011, Spector 2016 on the need for a restriction to excludable focus alternatives). We treat focus alternatives as syntactic objects derived by replacing focused expressions with other expressions of the same category and complexity, following Fox 1999, Katzir 2007, Fox & Katzir 2011. Nothing hinges on this choice, however.

(2) An integrated analysis of *only*:

$\llbracket \text{only}_C S \rrbracket$ is defined only if $\llbracket S \rrbracket = 1$.

If defined, $\llbracket \text{only}_C S \rrbracket = 1$ iff $\neg \exists S' \in C \cap \text{IE}(S): \llbracket S' \rrbracket = 1$.

(3) $\text{IE}(S) = \bigcap \{M \mid M \text{ is a maximal set in } F(S) \text{ s.t. } \{ \neg \llbracket \wedge S' \rrbracket \mid S' \in M \} \cup \{ \llbracket \wedge S \rrbracket \} \text{ is consistent} \}$

The sentence in (1) has the LF in (4), where *only* is adjoined to a clause. *Only* triggers the positive presupposition in (5): namely, the prejacent of *only* – that Gal was at the party – is true.

(4) $\llbracket \text{only}_C [\text{Gal}_F \text{ was at the party}] \rrbracket$

(5) Presupposition of (4): Gal was at the party

The exhaustive inference is derived from the relevant, excludable focus alternatives to the sister of *only*. The focus alternatives are listed in (6a). Those alternatives not entailed by the prejacent – specifically, those in which the subject denotes an individual distinct from Gal – are excludable and are given in (6b). Distinctness between two individuals is modeled as non-overlap, represented with ‘ $\otimes$ ’. The resulting exhaustive inference, given in (7), is that no individual distinct from Gal is relevant and was at the party. (That these two inferences pattern as presupposition and assertion, respectively, has been widely documented, albeit with some qualifications; see, e.g., Horn 1972, Beaver & Clark 2008, Tonhauser et al. 2013, Crnič 2024.)

(6) a. $F([\text{Gal}_F \text{ was at the party}]) = \{ [X \text{ was at the party}] \mid X \in F(\text{Gal}_F) \}$

b. $\text{IE}([\text{Gal}_F \text{ was at the party}]) = \{ [X \text{ was at the party}] \mid X \in F(\text{Gal}_F) \wedge \llbracket X \rrbracket \otimes \llbracket \text{Gal} \rrbracket \}$

(7) Assertion of (4): $\neg \exists S \in C \cap \text{IE}([\text{Gal}_F \text{ was at the party}]): \llbracket S \rrbracket = 1$

$$\Leftrightarrow \neg \exists x: x \otimes \text{Gal} \wedge x \text{ was at the party}$$

1.2 A distributed analysis of *only*

Another line of analysis is the distributed one, according to which the two inferences of *only* are generated by distinct operators (Crnič 2024, building on the treatment of exclusives as exceptives by von Stechow & Iatridou 2007; see also Horn 1992, McCawley 1993, van Rooy & Schulz 2007 for

related proposals). The first operator is *only* itself, defined in (8). It combines with a sentence and contributes the exhaustive inference – crucially, it does not contribute the positive presupposition.

(8) **A distributed analysis of *only* – the *only* component:**

$$\llbracket \text{only}_C S \rrbracket = 1 \text{ iff } \neg \exists S' \in C \cap \text{IE}(S): \llbracket S' \rrbracket = 1.$$

The second operator is identical to *only*, except that its exhaustive inference is presupposed. We label this operator *exh* (‘exhaustification’) and define it in (9) (following, esp., Bassi et al. 2021, Del Pinal et al. 2024). We assume that *exh* is present in sentences with *only* by definition, though its presence may be mandated by more general considerations not specific to *only*.¹ (For arguments that *exh* must presuppose the exhaustive inference in order to capture the behavior of *only*, see Crnič 2024. These align with independent arguments advanced in several other domains by Fox 2020, Bassi et al. 2021, Del Pinal et al. 2024, Guerrini & Wehbe 2025, etc. The reader is also referred to these papers for a more sophisticated characterization of *exh*.)

(9) **A distributed analysis of *only* – the *exh* component:**

$$\llbracket \text{exh}_C S \rrbracket \text{ is defined only if } \neg \exists S' \in C \cap \text{IE}(S): \llbracket S' \rrbracket = 1.$$

$$\text{If defined, } \llbracket \text{exh}_C S \rrbracket = 1 \text{ iff } \llbracket S \rrbracket = 1.$$

Unlike on the integrated analysis, the positive inference is thus not hardwired into the meaning of either *only* or *exh*. Instead, it arises from their interaction. To see how this works, consider the case of (1). The sentence has the structure in (10), with *exh* scoping over *only*, and with both operators associating with the same focused expression. (While multiple association is often restricted for other focus-sensitive expressions, see, e.g., Beck 2009, we assume it is possible in cases involving silent operators such as *exh*; see Crnič 2013, Bade & Sachs 2019 for discussion. For uniformity, we treat the alternatives to the sister of *exh* simply as focus alternatives.)

$$(10) \quad [\text{exh}_{C'} [\text{only}_C [\text{Gal}_F \text{ was at the party}]]]$$

The assertion of (10) corresponds to the meaning of the *only*-prefixed constituent, which is unaffected by *exh* and matches the assertive meaning of *only* on the integrated analysis:

$$\begin{aligned} (11) \quad \text{Assertion of (10): } & \llbracket [\text{only}_C [\text{Gal}_F \text{ was at the party}]] \rrbracket = 1 \\ & \Leftrightarrow \neg \exists S \in C \cap \text{IE}([\text{Gal}_F \text{ was at the party}]): \llbracket S \rrbracket = 1 \\ & \Leftrightarrow \neg \exists x: x \otimes \text{Gal} \wedge x \text{ was at the party} \end{aligned}$$

The presupposition of (10) is somewhat more complex. It says that no relevant and excludable alternative to $[\text{only}_C [\text{Gal}_F \text{ was at the party}]]$ is true – where ‘relevant’ means belonging to the resource domain C' of *exh*, and ‘excludable’ again reduces to the alternatives not entailed by the sister of *exh*, that is, alternatives of the form $[\text{only}_C [X_F \text{ was at the party}]]$, where X denotes a relevant individual distinct from *Gal*, (12). The reasoning is as follows: If the individual denoted by X overlaps with *Gal*, then *only* would negate fewer alternatives than it does when its associate is *Gal*; accordingly, the negation of such an alternative is incompatible with the sister of *exh*. By contrast, if X denotes an individual with no overlap with *Gal*, the corresponding alternative can be negated in unison with all other such alternatives, as we see in the computation in (13) below.

¹The distributed analysis of *only* has been informed by earlier work on connected exceptives, like *but Gal* in *No one but Gal was at the party*, which exhibit inferential patterns closely paralleling those of *only* (esp., Gajewski 2008, 2013, Hirsch 2016, Crnič 2018, all building on von Stechow 1993). We turn to issues involving exceptives in Sect. 5.

- (12) a. $F(S') = \{[\text{only}_C [X_F \text{ was at the party}]] \mid X \in F(\text{Gal}_F)\}$
 b. $IE(S') = \{[\text{only}_C [X_F \text{ was at the party}]] \mid X \in F(\text{Gal}_F) \wedge \llbracket X \rrbracket \otimes \text{Gal}\}$

Following Rooth 1992, Crnić 2024 assumes that the alternatives to proper names may also denote plural individuals. This fits naturally with the general characterization of alternatives outlined above, given the now standard view that proper names project full DP structures (see, e.g., Longobardi 1994, Geurts 1997, Borer 2005, Elbourne 2005, 2013, among others). Applying this to the exhaustive presupposition, we arrive at the interpretation that no individual distinct from Gal is such that no one distinct from them was at the party. Put differently: for every individual distinct from Gal, there is someone distinct from that individual who was at the party. This reasoning extends to the plural individual consisting of everyone except Gal; in that case, the presupposition entails that at least one member distinct from that plural individual was at the party, which yields that Gal was at the party. Because this inference entails all the others generated, the overall result is simply that Gal was at the party. This is the strong positive inference. (It should be noted that our above assumption about the alternatives to proper names is not crucial. More generally, the alternatives required to derive the strong positive presupposition – also for various other types of associates of *only* – are available independently, even without this assumption. See the Appendix.)

- (13) Presupposition of (10): $\neg \exists S \in C \cap IE(S'): \llbracket S' \rrbracket = 1$
 $\Leftrightarrow \neg \exists x: x \otimes \text{Gal} \wedge \neg \exists y: y \otimes x \wedge y \text{ was at the party}$
 $\Leftrightarrow \forall x: x \otimes \text{Gal} \rightarrow \exists y: y \otimes x \wedge y \text{ was at the party}$
 $\Leftrightarrow \text{Gal was at the party}$

1.3 An earlier argument for the distributed analysis

There is no difference between the meaning of (1) computed on the integrated analysis of *only*, provided in (5) and (7), and the meaning computed on the distributed analysis of *only*, provided in (11) and (13). Given this equivalence, is there any reason to favor one analysis over the other? At least one point of divergence has been identified: the distributed analysis predicts that, in principle, operators may intervene between *exh* and *only*. This possibility is schematized in (14). If such intervention occurs, it can affect the content of the positive presupposition.

- (14) **A distinguishing prediction (scope of operators):**
 A syntactic operator may, in principle, occur between *exh* and *only* at LF.
 Schematically: $[\text{exh}'_C [\textcircled{\text{OP}} [\text{only}_C \dots]]] \rightsquigarrow \text{Possible effect on meaning}$

This prediction was explored in Crnić 2024 and argued to be realized in some cases.

The data. A well-documented contrast arises with continuations of *only* sentences, as illustrated in (15). Specifically, the modal continuation in (15a) is noticeably more acceptable than the non-modalized continuation in (15b), even though both contradict the putative positive presupposition of the first sentence – namely, that Gal was at the party. This contrast was first noted by Horn 1972 and has since been discussed extensively (see, e.g., Horn 1992, 1996, Roberts 2006, 2011, van Rooy & Schulz 2007, Ippolito 2008, Beaver & Clark 2008, among others). By contrast, if a continuation contradicts the exclusive inference, it is judged unacceptable regardless of whether it is modalized, as shown in (16), where both continuations contradict the exclusive inference, namely, that no individual distinct from Gal was at the party (= that Tal was not at the party).

- (15) Among Gal and Tal, only Gal_F was at the party ...
 a. ... and maybe even she wasn't.
 b. #... and (in fact) even she wasn't.
- (16) Among Gal and Tal, only Gal_F was at the party ...
 a. #... and maybe even Tal was.
 b. #... and (in fact) even Tal was.

In addition, while the contrasts exemplified above are robust across speakers, speaker judgments tend to differ regarding the naturalness of the modalized continuation itself. This variation likewise requires an explanation.

Split scope. Crnić 2024 shows that the contrast between the continuations in (15), as well as the asymmetry between (15) and (16), are accounted for naturally on the distributed analysis of *only*. On this analysis, the first sentence in (15) may be parsed with *exh* taking scope above a covert assertion operator, ASSERT, as represented in (17) (see, e.g., Chierchia 2013, Meyer 2013, Cohen & Krifka 2014, Krifka 2014, Beck 2016, Fox 2016, Buccola & Haida 2019 on ASSERT and its scope interactions). The resulting LF has the interpretation in (18)-(19): the speaker considers it possible that Gal was at the party, and is committed to the belief that no one distinct from Gal was there (see Crnić 2024 for a detailed derivation).

- (17) [exh_{C'} [ASSERT [only_C [Gal_F was at the party]]]]
- (18) Presupposition of (17): $\Diamond_{sp}(\text{Gal was at the party})$
- (19) Assertion of (18): $\Box_{sp}(\neg \exists x: x \otimes \text{Gal} \wedge x \text{ was at the party})$

As required, this meaning is compatible with the modal continuation and incompatible with the non-modal continuation in (15), thereby accounting for the observed pattern. On the other hand, since the scope of *only* is fixed below ASSERT (as it cannot undergo covert movement above ASSERT, cf. Taglicht 1984), the exclusive inference is entailed by the sentence – hence, any continuation contradicting it results in infelicity. Furthermore, Crnić 2024 argues that the integrated analysis lacks an adequate account of the pattern, concluding that the data support the distributed analysis.²

Variation. Furthermore, the analysis provides a way to describe both (i) variation among speakers with respect to the extent to which they find the modalized continuation acceptable, as well as (ii) potential variation among occurrences of *only* phrases in different environments and contexts.

With respect to (i), which we signaled above, the variation can be captured by a parameter that tracks how readily speakers admit an intervening operator between *exh* and *only* (cf., e.g., Crnić et al. 2015, Singh et al. 2016 on potential cross-speaker variation in parsing preferences of exhaustified sentences). With respect to (ii), the distributed analysis is more expressive than the integrated one, and allows for the positive inference to differ, in principle, from the prejacent in many more cases than just the one described above. This expectation is not fully borne out (cf. Crnić 2024, Sect. 5). For example, while the epistemic modal sentence in (20) does admit a weak

²There are integrated analyses that discuss the pattern in (15)-(16) in detail (esp., Ippolito 2007, Beaver & Clark 2008). However, as argued in Crnić 2024, Sect. 4, these analyses face difficulties with projection facts and the strength of the positive presupposition, and ultimately fail to account for the pattern. See Sect. 4.2 below for a discussion of Beaver & Clark's 2008 approach in relation to the data under consideration in this paper.

positive presupposition along the lines observed above, the weaker reading is less readily available in certain other configurations, such as those in (21), as the editor and a reviewer have noted.

- (20) Only Gal must have been at the party and maybe even she wasn't.
 (21) a. ?We were required to only talk to Gal and were allowed to not even do that.
 b. ?Gal is certain to only talk to Ann and she may not even do that.

Constraints on the relation between *exh* and its associates across intervening operators have been observed, and in some cases derived, in a range of other domains (cf. Fox & Hackl 2006, Gajewski 2008, Chierchia 2013, Fălăuș 2014, Mayr 2014). The distributed analysis of *only* faces an analogous challenge, summarized in (22): it must provide both a comprehensive description of when split scope of *exh* and *only* is possible and an account of that description.

(22) **Dimensions of Variation (to-do):**

There are constraints – still to be specified – on splitting the scope of *exh* and *only*, pertaining to both speaker preferences and the nature of syntactic interveners.

Addressing this challenge should go hand in hand with the study of exceptives, which rely on the same underlying mechanisms and appear to display the same restricted behavior (cf. Crnić 2024). Rather than taking on this task, our aim in this paper is more modest: we present a different, independent argument in favor of the distributed analysis – an argument that we hope will help generate momentum to tackle this separate challenge above. Crucially, the force of our argument does not hinge on whether the two alternative-sensitive mechanisms can in fact be split at LF. (In the remainder of the paper, we set aside the modalized positive presupposition derived above, as it is orthogonal to the inferences primarily under discussion here.)

1.4 A preview – a new argument for the distributed analysis

The distributed analysis can also be distinguished from the integrated one in at least one other respect. While both analyses encode the exhaustive inference into the meaning of *only*, they diverge in how they derive the positive presupposition. On the integrated analysis, the positive presupposition is encoded directly in the lexical entry of *only*, alongside the exhaustive inference. By contrast, on the distributed analysis, the positive presupposition arises indirectly through the application of *exh*. Since *exh* quantifies over alternatives, and since its domain of quantification can, in principle, be contextually manipulated – just like that of *only* – this analysis introduces an extra layer of flexibility. Appropriate manipulations may result in presuppositions that differ from the canonical positive presupposition described above, as schematized in (23).

(23) **Another distinguishing prediction (domains of operators):**

The positive presupposition of *only* is derived indirectly on the distributed analysis and is, in principle, manipulable through the set of alternatives over which *exh* quantifies.

Schematically: [*exh*_C [*only*_C ...]] $\rightsquigarrow$ Possible effect on meaning

Manipulations of the sets of alternatives over which operators range are well-known to have interpretive consequences. One such consequence can be seen in the content of the exhaustive inference of *only*. Consider a context in which the question under discussion is who among Gal, Tal, and Sal was at the party. In this context, sentence (1) gives rise to the exhaustive inference in (24a), but not to those in (24b). The reason is that the domain of *only* is contextually restricted to alternatives of the form [*X_F was at the party*], where *X* is *Gal*, *Tal*, or *Sal*.

- (24) Q: Who among Gal, Tal and Sal was at the party?
 A: Only Gal_F was at the party.
 a. $\Rightarrow \neg(\text{Tal was at the party}) \wedge \neg(\text{Sal was at the party})$
 b. $\nRightarrow \neg(\text{Charlie was at the party}), \nRightarrow \neg(\text{Jimmy was at the party})$

Can a similar kind of context-dependence be identified in the case of *exh* when it accompanies *only*? This paper argues that the answer is ‘yes’. In Sect. 2, we review examples in which the positive presupposition of *only* surfaces as weaker than the strong inference derived above. Sect. 3 shows that both the data supporting a strong positive presupposition and that supporting a weaker one follow from the distributed analysis. Sect. 4 demonstrates that these data are problematic for integrated analyses. Sects. 5 and 6 discuss some challenges and conclude the paper.

2 Weakness and mutual exclusivity

Weak positive presupposition

There are vexing occurrences of *only* that seem to give rise to positive presuppositions weaker than the prejacent itself (see, e.g., Horn 1996, Geurts & van der Sandt 2004, Klinedinst 2005, Beaver & Clark 2008, Grosz 2012, Coppock & Beaver 2014, among others). One such case, due to Klinedinst 2005, is given in (25), where the focus is on *Cal State*. In ordinary contexts – where one graduates with a BA from a single institution – this sentence does not convey that Gal got her BA from Cal State, as we would expect if the prejacent of *only* were presupposed. Instead, the inference is merely that Gal got her BA from some institution other than Cal State. Similarly, sentence (26) does not convey that Gal is a grad student, but rather that she occupies some other position on the academic ladder (Beaver & Clark 2008). These inferences are also accompanied by a scalar inference that the prejacent ranks low among the focus alternatives, as indicated by the use of *at least* in some of their paraphrases; we return to this inference in a brief excursus below.

- (25) Gal didn’t only get her BA from Cal State_F.
Observed positive inference: Gal got her BA degree from somewhere
 (not: Gal got her BA from Cal State)
 (26) Gal isn’t only a grad student_F.
Observed positive inference: Gal is at least a grad student
 (not: Gal is a grad student)

Interrogative counterparts of the above sentences give rise to the same inferences, as exemplified in (27)-(28) – what projects in these sentences is again a weak positive presupposition that corresponds to existential quantification over the focus alternatives to the sister of *only*.

- (27) Did Gal only get her BA from Cal State_F?
Observed positive inference: Gal got her BA degree from somewhere
 (not: Gal got her BA from Cal State)
 (28) Is Gal only a grad student_F?
Observed positive inference: Gal is at least a grad student
 (not: Gal is a grad student)

It should be noted that the sentences in (26) and (28) also have a reading on which the prejacent of *only* is, in fact, presupposed, even in ordinary contexts. This holds, for example, if the pertinent question under discussion is whether Gal is a grad student or whether she also performs some non-academic activity. Such a reading is less readily accessible for the sentences in (25) and (27) in ordinary contexts, unless we assume that focus falls on the whole VP; that is, the sentences in (25) and (27) cannot be as easily construed to have alternatives compatible with Gal getting a BA from Cal State in ordinary contexts.

Sufficiency modal constructions

A merely existential positive presupposition has also been argued to arise in the notoriously challenging sufficiency modal constructions with *only* (see, e.g., von Fintel & Iatridou 2005, 2007, Krasikova & Zhechev 2006, Enguehard 2021, Alonso-Ovalle & Hirsch 2022, Condoravdi & Francez 2022, among others). Consider sentence (29). The sister of *only* in (29) corresponds *prima facie* to your having to go to the North End to get good cheese. But this is clearly not an entailment of the sentence. Instead, the positive inference of (29) can be captured with a paraphrase involving existential quantification: you have to go somewhere to get good cheese.

(29) To get good cheese, you only have to go to the North End_F.

Observed positive inference: You have to go somewhere
(not: You have to go to the North End)

Generalization

Klinedinst 2005 submits that *only* sentences give rise to existential positive presuppositions just in case the sister of *only* and its relevant focus alternatives are mutually incompatible in the context (for example, getting a BA from one institution is contextually incompatible with getting a BA from a different one; being a grad student is contextually incompatible with being a postdoc or a professor, etc). This is summarized in (30).

(30) **Weak and Strong Content:**

An *only* sentence induces a weak positive presupposition if the focus alternatives to the sister of *only* are mutually incompatible in the context. Otherwise the sentence induces a strong positive presupposition, which corresponds to the prejacent of *only*.

The generalization extends naturally to sufficiency modal constructions. This becomes clear by leveraging an insight from Condoravdi & Francez 2022 regarding their restricted distribution (see also von Fintel & Iatridou 2005, 2007 for extensive discussion of these restrictions). Condoravdi & Francez argue that the modals that participate in sufficiency modal constructions are precisely those that privilege effort minimization (e.g., teleological modals). How does this connect to the generalization in (30)? On the assumption that the worlds quantified over by the modal include only those in which minimal effort to get good cheese is exerted, the focus alternatives to the sister of *only* in sentence (29), provided in (31), are mutually incompatible: the effort of going to one place is distinct from the effort of going somewhere else farther away. By contrast, if no such minimization requirement is imposed, as with epistemic modals, the generalization cannot be met, and sufficiency modal readings do not arise (unless mutual incompatibility among the focus alternatives is secured in some other way by the context).

(31) a. To get good cheese, you have to go to the North End.

- b. To get good cheese, you have to go to New York City.
- c. To get good cheese, you have to go to France.

The goal of this paper is to re-assess the consequences of the availability of weak positive inferences for the analysis of *only*. More specifically, we contend that these observations favor the distributed analysis over the more traditional integrated ones.

Excursus: scalar inferences

Before proceeding to study the consequences of the availability of weak positive inferences, we bracket out a scalar inference – or perhaps multiple scalar inferences – that accompany *only* and that we pointed out in the preceding examples (see, e.g., Jacobs 1983, Klinedinst 2005, Beaver & Clark 2008, Grosz 2012, Zeevat 2013, Coppock & Beaver 2014, Alxatib 2020, Greenberg 2022, von Fintel 2024, among others). This inference, or these inferences, entail that the prejacent is ranked lower on some salient scale than the other relevant focus alternatives to the sentence. That such an inference is indeed generated is further demonstrated in (32), an example drawn from von Fintel & Iatridou 2005: the sentence is perceived as facetious since curing cancer is conveyed to be ranked lower than other relevant alternatives, perhaps even conveyed to be unremarkable (whether these are separate inferences of *only* is discussed in Greenberg 2022).

(32) %To win the Nobel Prize, you only have to cure cancer.

The reason why we set the scalar inference aside for much of this paper is that it appears to be independent of the weak positive inference. We can observe it even in *only*-sentences where the relevant alternatives are mutually compatible, as in (32) (see Sect. 4.2 for further illustration). There are several ways in which the scalar inference could be, and has been, incorporated into the analysis of *only*: one approach is to treat it as a separate presupposition encoded in the lexical entry of *only* (e.g., Klinedinst 2005, Grosz 2012, Greenberg 2022); another is to derive it from a mechanism external to *only* (cf. Panizza & Sudo 2020). For our purposes, all that matters is that each of these approaches is, in principle, compatible with the distributed analysis. (See Sect. 4.2 for a detailed discussion of an approach on which scalar ordering features centrally in the generation of the positive presupposition, due to Klinedinst 2005, Beaver & Clark 2008.)

3 Distributed resolution

On the distributed analysis of *only*, positive presuppositions are derived by means of exhaustification. In particular, the strong positive presupposition was shown to follow from *exh* negating all relevant excludable alternatives. Manipulation of the alternatives *exh* ranges over, to the extent it is possible, may affect the positive presupposition. We argue that such manipulation is possible and show how it accounts for the variability in the strength of positive inferences.

3.1 Exhaustification and pruning

The alternatives that *exh* and *only* range over are co-determined by the structure of their sisters and the context. We have implemented dependence on context by equipping each quantifier with a silent resource domain that is subscripted in our representations and resolved by the context (e.g., von Fintel 1994). In line with our earlier example, when considering response A in (33), exhaustification naturally ranges over alternatives involving Gal, Tal, and Sal.

- (33) Q: Who among Gal, Tal and Sal was at the party?
 A: Gal_F was at the party.
- a. $\Rightarrow \neg(\text{Tal was at the party}) \wedge \neg(\text{Sal was at the party})$
 - b. $\nRightarrow \neg(\text{Charlie was at the party}), \nRightarrow \neg(\text{Jimmy was at the party})$

More specifically, the response to the question in (24) is parsed as in (34a), where the resource domain of *exh* is resolved to contain alternatives conveying that Gal, Tal and Sal were at the party, (34b). All the alternatives distinct from the sister of *exh* are excludable, as provided in (34c), and their exclusion gives us the exhaustified meaning in (35). The exhaustive inference is presupposed and can be accommodated if it is not already part of the common ground (see, again, Fox 2020, Bassi et al. 2021, Del Pinal et al. 2024, Guerrini & Wehbe 2025 for arguments that the inference generated by *exh* should be analyzed as a presupposition).

- (34) a. $[\text{exh}_C [\text{S Gal}_F \text{ was at the party}]]$
 b. $C \cap F(S) = \{[X \text{ was at the party}] \mid \llbracket X \rrbracket \text{ is Gal, Tal or Sal}\}$
 c. $\text{IE}(S) = \{[X \text{ was at the party}] \mid X \in F(\text{Gal}_F) \wedge \llbracket X \rrbracket \otimes \text{Gal}\}$
- (35) Gal was at the party $\wedge \neg \exists S' \in C \cap \text{IE}(S): \llbracket S' \rrbracket = 1$
 $\Leftrightarrow$ Gal was at the party $\wedge \neg \exists z: z \otimes \text{Gal} \wedge z \text{ was at the party}$

It follows from the definition of *exh* that restricting the alternative set via the resource domain affects only excludable alternatives (see Katzir 2014 for a review of supporting arguments). At the same time, there is evidence that such restrictions are subject to further constraints, and different proposals have been put forward concerning what kinds of restrictions are permissible. We return to these constraints in Sect. 3.3. Before doing so, however, we show how pruning of alternatives can account for the observed variability in the strength of the positive presupposition of *only*.

3.2 Weakening through pruning

In our treatment of the sentence *Only Gal_F was at the party*, whose LF is repeated below, we took *exh* to quantify over all focus alternatives to its sister, as explicitly stated in (37) – that is, the contribution of the domain restriction of *exh*, C' , was effectively vacuous.

- (10) $[\text{exh}_{C'} [\text{only}_C [\text{Gal}_F \text{ was at the party}]]]$
- (36) Implicit assumption in the preceding text (yielding the strong positive presupposition):
 $F([\text{only}_C [\text{Gal}_F \text{ was at the party}]]) \subseteq C'$

The puzzle

If the same kind of vacuous domain restriction described above obtains in the sentence *Gal is only a grad student_F*, whose LF is in (37), the sentence should be infelicitous. Let us see why.

- (37) $[\text{exh}_{C'} [\text{S'} \text{ only}_C [\text{S Gal is a } [\text{grad student}]_F]]]$

The focus alternatives to the sister of *only* in (37) are provided in (38a), while the excludable alternatives among these are characterized in (38b). The excludable alternatives include, among others, all focus alternatives whose negation is contextually entailed by the sister of *only*. But, for example, they do not include the alternative $[\text{Gal is an introvert}_F]$, as it is not in all maximal sets

of alternatives that can be jointly negated consistently with the sentence being true (namely, it is not contained in the set that contains $[Gal \text{ is an } extrovert_F]$). (For concreteness, we single out the noun *non-academic* as an alternative; other nouns would do just as well. See also the Appendix.)

- (38) a. $F(S) = \{[Gal \text{ is a } NP_F] \mid NP \in F([grad \text{ student}]_F)\}$
b. $\{[Gal \text{ is a } NP_F] \mid NP \in \{postdoc, professor, non-academic\}\} \subseteq IE(S)$

The focus alternatives to the sister of *exh* in (37) contain the operator *only_C* and are provided in (39a). The set of excludable alternatives is given in (39b).

- (39) a. $F(S') = \{[only_C [Gal \text{ is a } NP_F]] \mid NP \in F([grad \text{ student}]_F)\}$
b. $\{[only_C [Gal \text{ is a } NP_F]] \mid NP \in \{postdoc, professor, non-academic\}\} \subseteq IE(S')$

On the assumption that the focus alternatives pertaining to positions on the academic ladder are contained in the resource domain of *only*, and that all the focus alternatives are contained in the resource domain of *exh*, we obtain the assertion in (40): Gal is not a postdoc or a professor.

- (40) Assertion of (37) (all focus alternatives are relevant):
 $\neg \exists S \in C \cap IE([only_C [Gal \text{ is a } [grad \text{ student}]_F]]): \llbracket S \rrbracket = 1$
 $\Leftrightarrow \neg(Gal \text{ is a postdoc}) \wedge \neg(Gal \text{ is a professor})$

The presupposition corresponds to Gal being a grad student. More concretely, the negation of the excludable alternatives yields the conjunction of the following propositions: that Gal is a grad student or a professor, that Gal is a grad student or a postdoc, and that Gal is a grad student, a postdoc, or a professor. This conjunction is contextually equivalent to the simple statement that Gal is a grad student (since one cannot be both a professor and a postdoc). The derivation is represented step by step in (41). (See the Appendix for a fuller set of alternatives.)

- (41) Presupposition of (37) :
 $\neg \exists S \in C' \cap IE([only_C [G \text{ is a } [grad \text{ student}]_F]]): \llbracket S \rrbracket = 1$
 $\Leftrightarrow \neg \llbracket [only_C [Gal \text{ is a postdoc}_F]] \rrbracket \wedge \neg \llbracket [only_C [Gal \text{ is a professor}_F]] \rrbracket \wedge$
 $\neg \llbracket [only_C [Gal \text{ is a non-academic}_F]] \rrbracket$
 $\Leftrightarrow (Gal \text{ is a grad student} \vee Gal \text{ is a professor}) \wedge$
 $(Gal \text{ is a grad student} \vee Gal \text{ is a postdoc}) \wedge$
 $(Gal \text{ is a grad student} \vee Gal \text{ is a postdoc} \vee Gal \text{ is a professor})$
 $\Leftrightarrow_c Gal \text{ is a grad student}$

On this analysis, however, the sentence is predicted to be unassertable in ordinary contexts, since its assertion would violate a basic condition on assertability – namely, that assertions must be informative, as stated in (42) (esp., Stalnaker 1974, 1978). In particular, structure (37), on the resolution of resource domains we are entertaining, triggers a presupposition that contextually entails the assertion: Gal being a grad student entails that she is not a postdoc, a professor, or a non-academic. Accordingly, the sentence should be infelicitous, contrary to fact. This is a conundrum.

- (42) **Informativity Constraint:**

No LF may have an assertive meaning contextually entailed by its presupposition.

The decision to include all focus alternatives in the resource domain of *exh* is arbitrary, however. The resource domain can be resolved, all else equal, to any subset of the focus alternatives to

the sister of *exh*. Accordingly, cooperative conversational participants can avoid the conundrum by resolving the domain of *exh* differently. Instead of letting *exh* range over the full set of excludable alternatives, they can restrict its domain so that the presupposition no longer entails the assertion. In what follows, we illustrate how such an alternative resolution of the domain of *exh* yields a felicitous outcome in cases like (37).

A resolution via pruning of alternatives

The domain of *exh* in (37) need not contain all excludable alternatives. In particular, it may contain just the excludable alternative built on *non-academic* – that is, other focus alternatives can be ‘pruned’. This is represented in (43). Since *exh* lets the meaning of its sister through at the level of assertion, this restriction does not affect the asserted meaning of (37), as shown in (44).

(43) Domain resolutions in (37):

$$\begin{aligned} C \cap F(S) &= \{[\text{Gal is a NP}_F] \mid \text{NP} \in \{\text{grad student, postdoc, professor}\}\} \\ C' \cap \text{IE}(S') &= \{[\text{only}_C [\text{Gal is a non-academic}_F]]\} \end{aligned}$$

(44) Assertion of (37) on the resolution in (43):

$$\begin{aligned} \neg \exists S \in C \cap \text{IE}([\text{Gal is a [grad student]}_F]): \llbracket S \rrbracket &= 1 \\ \Leftrightarrow \neg(\text{Gal is a postdoc}) \wedge \neg(\text{Gal is a professor}) \end{aligned}$$

Turning now to the exhaustification of (37), the domain resolution in (43) leads to a weak positive presupposition. Namely, the sole excludable alternative in the resource domain of *exh* – $[\text{only}_C [\text{Gal is a non-academic}_F]]$ – has the meaning in which the alternatives built on *grad student*, *postdoc* and *professor* are all excludable and excluded, as provided in (45).

(45) $\llbracket [\text{only}_C [\text{Gal is a non-academic}_F]] \rrbracket =$

$$\begin{aligned} \neg \exists S \in C \cap \text{IE}([\text{Gal is a non-academic}_F]): \llbracket S \rrbracket &= 1 \\ \Leftrightarrow \neg(\text{Gal is a grad student}) \wedge \neg(\text{Gal is a postdoc}) \wedge \neg(\text{Gal is a professor}) \end{aligned}$$

The presupposition generated by exhaustification is, accordingly, the negation of this meaning: the inference that Gal is a grad student, a postdoc, or a professor. We have thus derived a meaning that does not violate the Informativity Constraint, as desired: the presupposition that Gal is an academic (a grad student, postdoc, or professor), (46), does not contextually entail the assertion that she is not a postdoc or a professor, (44). Moreover, the presupposition and the assertion jointly entail that Gal is a grad student, the observed inference of the sentence.

(46) Presupposition of (37) on the resolution (43):

$$\begin{aligned} \neg \exists S \in C' \cap \text{IE}([\text{only}_C [\text{Gal is a [grad student]}_F]]): \llbracket S \rrbracket &= 1 \\ \Leftrightarrow \llbracket [\text{only}_C [\text{Gal is a non-academic}_F]] \rrbracket &= 0 \\ \Leftrightarrow (\text{Gal is a grad student} \vee \text{Gal is a postdoc} \vee \text{Gal is a professor}) \end{aligned}$$

If we embed the sentence on this resolution under negation or in a question, we straightforwardly derive the weak positive presupposition observed in such examples – namely, that Gal is a grad student, a postdoc, or a professor. For example, the negated sentence in (26), repeated below, can be parsed as in (47). (If *exh* were to take scope above negation, the result would be pathological; see Crnič 2024, Sect. 3, for discussion.)

(26) Gal isn't only a grad student_F.

Observed positive inference: Gal is at least a grad student
(not: Gal is a grad student)

(47) [neg [exh_{C'} [only_C [Gal is a [grad student]_F]]]]

The meaning of (47) is provided in (48)-(49): the presupposition is identical to that of the unembedded sentence, (48), while the assertion is simply the negation of the asserted meaning of the unembedded sentence, (49). Together, the presupposition and assertion entail that Gal is either a postdoc or a professor, as desired. The meaning is not in violation of the Informativity Constraint: Gal being an academic does not entail that she is either a postdoc or a professor. The same reasoning extends to other embedded occurrences of the sentence.

(48) Presupposition of (47) on the resolution (43):

(Gal is a grad student $\vee$ Gal is a postdoc $\vee$ Gal is a professor)

(49) Assertion of (47) on the resolution (43):

(Gal is a postdoc $\vee$ Gal is a professor)

Summary

The distributed analysis provides a straightforward account of how both strong and weak positive inferences can be derived: they result from different resolutions of the domain of *exh*. When no pruning applies, a strong positive presupposition is generated; when appropriate pruning does apply, a weak positive presupposition emerges. This derives the cases in which we observe weak positive presuppositions. Concerns about possible overgeneration loom large, however.³

3.3 Pruning and its limits

The derivation of a weak positive presupposition through pruning, as described above, overgenerates. Most notably, it incorrectly predicts weak positive presuppositions to be possible for all *only* sentences. This issue arises because pruning has been left completely unchecked in the preceding discussion. Parallel issues have been recognized elsewhere in the theory of exhaustification, where independent constraints on pruning have been proposed. We show that these constraints carry over naturally and correctly restrict the overgeneration in the distributed analysis of *only*.

Overgeneration, all else equal

Overgeneration with respect to the variability in the strength of positive inferences can be illustrated with a sentence like *Not only Gal_F was at the party*. The sentence gives rise to the strong positive presupposition that Gal was at the party, as stated in (50). Unlike in the case of *Gal is (not) only a grad student_F*, however, the strong positive presupposition here does not violate the

³Before we turn to some genuine domain resolution candidates, let us dismiss a clearly defective one. Namely, we could prune all the alternatives from the domain of *exh* – that is, we could assume that *C'* in our examples above picks out an empty set. This would make the contribution of *exh* vacuous. Although such a resolution might be possible elsewhere, given that one does not always compute exhaustified meanings of sentences (cf. Grice 1975), it is impossible with *only*. We can attribute this to the combination of the core assumptions of the distributed analysis – that *exh* obligatorily accompanies *only* and that it associates with the same focused expression as *only* – and an independent assumption from the exhaustification literature – that *exh* cannot be vacuous (e.g., Fox & Spector 2009, 2018, Spector 2014, among others; see also Sect. 5.1 below).

Informativity Constraint: Gal having been at the party – which is what the sentence presupposes – does not entail that someone else was – which is what the sentence asserts.

(50) Not only Gal_F was at the party.

Observed positive presupposition: Gal was at the party

However, judicious pruning along the lines described above can lead to an interpretation on which Gal need not have been at the party – that is, there is a parse of the sentence that yields merely a weak positive presupposition that some relevant individual was at the party. More concretely, this can be achieved if the domain of *exh* in (51) is resolved to contain merely one excludable alternative, say, $[only_C [Charlie_F \text{ was at the party}]]$, as provided in (52):

(51) $[neg [exh_{C'} [S' \text{ only}_C [S \text{ Gal}_F \text{ was at the party}]]]]$

(52) Domain resolution in (51):

- a. $C \cap F(S) = \{[Gal_F \text{ was at the party}], [Tal_F \text{ was at the party}], [Sal_F \text{ was at the party}]\}$
- b. $C' \cap IE(S') = \{[only_C [Charlie_F \text{ was at the party}]]\}$

The meaning of the sole excludable alternative in the domain of *exh* is provided in (54), where the alternatives built on *Gal*, *Tal* and *Sal* make up the domain of *only* and are excludable and excluded.

(53) $C \cap IE([Charlie_F \text{ was at the party}]) = \{[X_F \text{ was at the party}] \mid X \in \{Gal, Tal, Sal\}\}$

(54) $[[[only_C [Charlie_F \text{ was at the party}]]]] =$
 $\neg(Gal \text{ was at the party}) \wedge \neg(Tal \text{ was at the party}) \wedge \neg(Sal \text{ was at the party})$

The full interpretation of the sentence on this resolution is, accordingly, provided in (55)-(56): it asserts that either Tal or Sal was at the party, and presupposes that Gal, Tal, or Sal was at the party. These inferences can be jointly true if, say, only Tal was at the party. But since the sentence in actual fact gives rise to only a strong positive inference – namely, that Gal was at the party – the proposal as currently formulated overgenerates (see Roberts 2011 for discussion of this issue).

(55) Assertion of (51) on the resolution in (52):

$$\neg(\neg\exists S' \in C \cap IE(S): \llbracket S' \rrbracket = 1)$$

$$\Leftrightarrow (Tal \text{ was at the party} \vee Sal \text{ was at the party})$$

(56) Presupposition of (51) on the resolution in (52):

$$\neg\exists S \in C' \cap IE(S'): \llbracket S \rrbracket = 1$$

$$\Leftrightarrow \llbracket [only_C [Charlie_F \text{ was at the party}]] \rrbracket = 0$$

$$\Leftrightarrow (Gal \text{ was at the party} \vee Tal \text{ was at the party} \vee Sal \text{ was at the party})$$

An overly weak positive presupposition is generated because the current setup incorrectly permits arbitrary gerrymandering of the domain of *exh*. In particular, while narrow domain restrictions of the kind we described are possible in cases where weak positive presuppositions are observed, they are not permitted in cases where strong positive presuppositions are observed. A descriptive constraint that curtails this overgeneration is given in (57) – it is a transposition of Klinedinst's generalization into the new framework: alternatives may be pruned in the cases at hand if and only if doing so is necessary to enable a felicitous assertion, specifically, to avoid a violation of the Informativity Constraint. In sentences such as *(Not) only Gal_F was at the party*,

where a strong positive presupposition does not give rise to a violation of the Informativity Constraint, pruning is not permitted.

(57) **Descriptive constraint:**

Exhaustification that accompanies *only* must range over all the focus alternatives to its sister unless such a resolution violates the Informativity Constraint.

Does this *ad hoc* constraint follow from any independent principles?

Not all else is equal – constraints on pruning

A variety of constraints on pruning of alternatives from the domain of *exh* have been put forward, typically motivated by intricate argumentation that we cannot reproduce here (e.g., Fox & Katzir 2011, Katzir 2014, Crnič et al. 2015, Trinh & Haida 2015, Trinh 2019, Bar-Lev 2024, and others). For our purposes, it suffices to adopt a constraint argued for by Crnič et al. 2015 and Bar-Lev 2024: any pruning of alternatives must lead to a reduction in the information conveyed by the exhaustified sentence. In our framework, where the exhaustive inference of *exh* is presupposed, a parse of a sentence conveys less information than another if its assertive meaning, together with its exhaustive presupposition, is entailed by that of the other parse. Equivalently – and innocuously, given that exhaustive inferences are easily accommodated (Bassi et al. 2021) – one parse conveys less information than another if and only if the accommodated meaning of the former is entailed by the accommodated meaning of the latter. The condition employing the standard accommodation operator is stated in (58) (cf. Beaver & Krahmer 2001):⁴

(58) **Weakening Constraint:**

$$\begin{aligned} & *[\text{exh}_C S] \text{ if there exists } C' \text{ such that } C \subset C' \text{ and} \\ & \text{ACCOM}(\llbracket [\text{exh}_C S] \rrbracket) \Rightarrow \text{ACCOM}(\llbracket [\text{exh}_{C'} S] \rrbracket). \\ & \text{(where } \text{ACCOM}(p) = 1 \text{ if } p = 1; \text{ otherwise, } \text{ACCOM}(p) = 0) \end{aligned}$$

The constraint in (58) correctly blocks the weakening of the positive presupposition in sentences such as *Not only Gal_F was at the party*. This is because the information conveyed by exhaustification – that is, the information conveyed by $[\text{exh}_{C'} [S' \text{ only}_C [S \text{ Gal}_F \text{ was at the party}]]]$ – without pruning is equivalent to that conveyed with the proposed pruning. In both cases, the asserted meaning – which is unaffected by the choice of the resource domain of *exh* – combined with either the weak or strong presupposition yields the inference that Gal was at the party. This is illustrated in (59): the top part of (59) presents the (accommodated) meaning under exhaustification without pruning, while the bottom part of (59) presents the result of exhaustification with the pruned domain resolution in (52).

$$\begin{aligned} (59) \quad & \text{ACCOM}(\llbracket [\text{exh}_{C'} [S' \text{ only}_C [S \text{ Gal}_F \text{ was at the party}]]] \rrbracket) \text{ without pruning:} \\ & \neg(\text{Tal was at the party}) \wedge \neg(\text{Sal was at the party}) \wedge (\text{Gal was at the party}) \end{aligned}$$

⁴As the original formulation of the constraint in Crnič et al. 2015, p. 283, was designed for a non-presuppositional version of *exh*, it is simpler in that it does not need to employ accommodation. We reproduce it in (i), with the superscript NP indicating that the exhaustive inference is not presupposed but rather conjoined to the meaning of the sister of *exh* at the assertive level.

(i) **Weakening Constraint (original):**

$$*[\text{exh}_C^{NP} S] \text{ if there exists } C' \text{ such that } C \subset C' \text{ and } \llbracket [\text{exh}_C S] \rrbracket \Rightarrow \llbracket [\text{exh}_{C'} S] \rrbracket.$$

$\Leftrightarrow$

ACCOM($\llbracket \llbracket \text{exh}_{C'} [\text{only}_C [\text{Gal}_F \text{ was at the party}]] \rrbracket \rrbracket$) with pruning (resolution in (52)):
 $\neg(\text{Tal was at the party}) \wedge \neg(\text{Sal was at the party}) \wedge$
 $(\text{Gal was at the party} \vee \text{Tal was at the party})$

However, by the same logic, the Weakening Constraint should also block pruning in sentences like *Gal is only a grad student_F*, that is, in sentences in which we argued that pruning obtains. Unlike in the example above, though, there are compelling reasons to override the constraint here: pruning prevents a violation of the Informativity Constraint, repeated below. Accordingly, to capture the observed distribution of weak positive presuppositions, the Weakening Constraint must be defeasible and ranked below the Informativity Constraint, as stated in (60).

(42) **Informativity Constraint:**

No LF may have an assertive meaning contextually entailed by its presupposition.

(60) **Constraint Ranking:**

Informativity Constraint $\gg$ Weakening Constraint

The complex state of affairs that we set out to explain falls out as follows: sentences in which the sister of *only* has mutually compatible focus alternatives give rise to strong positive presuppositions (they satisfy the Informativity Constraint, so pruning is blocked by the Weakening Constraint); sentences in which the sister of *only* has (contextually) mutually incompatible focus alternatives give rise to weak positive presuppositions (pruning is admitted in order to satisfy the Informativity Constraint). The descriptive constraint on the distribution of weak positive presuppositions in (57) is thus a consequence of these extant, independently motivated constraints on pruning and assertion if the latter outrank the former.

Summary

What positive inferences an *only* sentence gives rise to is conditioned by independent constraints on pruning, specifically, the Weakening Constraint. The Weakening Constraint may be bypassed if doing so is necessary to avoid a pathologically uninformative sentence. A summary of the admitted derivations is provided in Tables 1 and 2.

	Resolution	Informativity	Weakening	Positive inf.
✗	No pruning of focus alternatives in C'	*		strong
☞	Pruning of all but the strongest focus alternatives in C'		*	weak

Table 1: Predicted inferences for sentences with contextually mutually incompatible focus alternatives, such as $\llbracket \text{exh}_{C'} [\text{only}_C [\text{Gal is a } [\text{grad student}]_F]] \rrbracket$.

The discussion raises the question of whether the constraints on pruning, one of which is defeasible, adequately capture the full distribution of exhaustified readings. As always, this question is difficult to answer, especially in the long run. Nonetheless, introducing the defeasibility of the Weakening Constraint does not affect the cases for which it was originally proposed (since in none of those does the Informativity Constraint come into play; see Crnič et al. 2015, Bar-Lev 2024 for discussion). Moreover, the ranking of the constraints does not have undesirable repercussions in

	Resolution	Informativity	Weakening	Positive inf.
☞	No pruning of focus alternatives in C'			strong
✗	Pruning of all but the strongest focus alternatives in C'		*	weak

Table 2: Predicted inferences for sentences with contextually mutually incompatible focus alternatives, such as $[exh_{C'} [only_C [Gal_F \text{ was at the party}]]]$.

other cases in which exhaustification is obligatory and responsible for deriving the restricted distribution of certain expressions, as we demonstrate on the case of exceptives in Sect. 5.1.

Excursus: Intermediate presuppositions and additional constraints

Further constraints on pruning have been recognized, specifically those pertaining to the nature of contextual relevance (esp., Groenendijk & Stokhof 1984, Lewis 1988, Fox & Katzir 2011). These constraints condition the structure of licit domain restrictions; they primarily involve the closure of relevance under Boolean operations and under contextual equivalence, as stated in (61):

- (61) **Relevance in context in c :**
- a. If S is relevant in c , then $\neg S$ is relevant in c .
 - b. If S and S' are relevant in c , then $S \vee S'$ and $S \wedge S'$ are relevant in c .
 - c. If S is relevant in c , then any S' is relevant in c if $S \Leftrightarrow_c S'$.

While we adopt these constraints – and even assume they are indefeasible (the updated full ranking of the constraints is provided in (62)) – their role in our derivations is minor and difficult to detect; hence, we set them aside in subsequent derivations for brevity.

- (62) **Constraint Ranking (updated):**
 Informativity Constraint, Relevance $\gg$ Weakening Constraint

Specifically, the further constraint rules out certain derivations that match ours with respect to the accommodated meanings of the sentences under discussion, but diverge with respect to how these meanings are divided between assertion and presupposition. For example, consider the sentence *Gal is only a grad student_F*, with the parse in (63). Let the domain restrictions be as in (64), where the alternative built on *postdoc* is pruned from the domain of *exh* (the assertion – that is, the meaning of the sister of *exh* – is relevant by default; e.g., Magri 2009, 2011).

- (63) $[exh_{C'} [_{S'} only_C [{}_S Gal \text{ is a } [grad \text{ student}]_F]]]$

- (64) **Domain resolution in (63):**
- a. $C \cap F(S) = \{[Gal \text{ is a } NP_F] \mid NP \in \{grad \text{ student}, postdoc, professor\}\}$
 - b. $C' \cap F(S') = \{[only_C [Gal \text{ is a grad student}_F]], [only_C [Gal \text{ is a professor}_F]]\}$

The meaning of the sentence on this selection of domains of alternatives is provided in (65)-(66). The presupposition is stronger than what we derived above: it is not merely that Gal has some academic position, but that she is either a grad student or a postdoc. The parse is admitted on the initial ranking of constraints as pruning is needed to avoid a violation of the Informativity Constraint, as discussed above.

- (65) Assertion of (63) on the resolution (64):
 $\neg(\text{Gal is a postdoc}) \wedge \neg(\text{Gal is a professor})$
- (66) Presupposition of (63) on the resolution (64):
 $(\text{Gal is a grad student} \vee \text{Gal is a postdoc})$

The ranking in (62), however, does not admit the selection of domains in (64). Specifically, in contexts under consideration, the set in (67) is not closed under Boolean operations: if C' contains the alternatives listed in (64), then it must also contain the alternative $[\text{only}_C [\text{Gal is a postdoc}_F]]$.⁵

(67) **Consequence of the constraints on relevance for the preceding:**

If $\{[\text{only}_C [\text{Gal is a grad student}_F]], [\text{only}_C [\text{Gal is a professor}_F]]\} \subseteq C'$,
then it must also hold that $\{[\text{only}_C [\text{Gal is a postdoc}_F]]\} \subseteq C'$.

Crucially, the consequence derived above does not extend to the domain restrictions discussed earlier, namely, that the domain of *exh* consists of the alternatives in (68). In this case, the constraint on relevance requires no other alternative to be in the domain of *exh*; specifically, no other focus alternative can be derived from the Boolean combination of the alternatives in C' .

- (68) a. $C \cap F(S) = \{[\text{Gal is a NP}_F] \mid \text{NP} \in \{\text{grad student, postdoc, professor}\}\}$
b. $C' \cap F(S') = \{[\text{only}_C [\text{Gal is a grad student}_F]], [\text{only}_C [\text{Gal is a non-academic}_F]]\}$

Again, we leave Relevance out of our subsequent discussion, solely for brevity. It can be easily checked that the constraint is satisfied in all the parses and computations to come.

3.4 Sufficiency modal constructions

Sufficiency modal constructions, repeated below, have been notoriously challenging to analyze. Particularly relevant to our discussion is the observation that a sentence like (29) conveys that it suffices to go to the North End to get good cheese, rather than that it is necessary to do so (see, e.g., von Stechow & Iatridou 2007, Alonso-Ovalle & Hirsch 2022 for comprehensive analyses).

- (29) To get good cheese, you only have to go to the North End_F.

Observed positive inference: You have to go somewhere
(not: You have to go to the North End)

⁵If the alternatives in (64) are relevant, then so is their conjunction and all the sentences equivalent to it:

- (i) $([[[\text{only}_C [\text{G. is a grad student}_F]]] \wedge [[[\text{only}_C [\text{G. is a professor}_F]]]]) \Leftrightarrow_c [[[\text{only}_C [\text{G. is a non-academic}_F]]]]$

From this it follows that sentences *Gal is a grad student/professor* are relevant as well:

- (ii) a. $(\neg[[[\text{only}_C [\text{G. is a non-academic}_F]]] \wedge [[[\text{only}_C [\text{G. is a grad student}_F]]]]) \Leftrightarrow_c [[[\text{G. is a grad student}]]]$
b. $(\neg[[[\text{only}_C [\text{Gal is a non-academic}_F]]] \wedge [[[\text{only}_C [\text{Gal is a professor}_F]]]]) \Leftrightarrow_c [[[\text{Gal is a professor}]]]$

Finally, since the conjunction of the negations of these sentences is relevant, and since this is equivalent to the focus alternative $[\text{only}_C [\text{Gal is a postdoc}_F]]$, this latter alternative must be relevant too, as stated in the main text:

- (iii) $(\neg[[[\text{Gal is a grad student}]]] \wedge \neg[[[\text{Gal is a professor}]]]) \Leftrightarrow_c [[[\text{only}_C [\text{Gal is a postdoc}_F]]]]$

The puzzle and its resolution

While the observed positive inference in (29) is weaker than what the strong positive inference of the sentence would be, it differs from the weak positive inferences discussed above and cannot be derived completely analogously. Namely, if a weak positive inference is derived along the lines we have done above, we still get an inference that there exists a destination to which you have to go to get good cheese, contrary to fact. To address this issue, von Fintel & Iatridou 2007 propose, first, to split the negative quantifier encoded in *only* into a negation and an exceptive existential quantifier, and second, to interpret the latter within the scope of the modal in sentences like (29). We reproduce their proposal on the distributed analysis of *only*. The assumptions concerning the morphology of *only* are given in (69), broadly following von Fintel & Iatridou 2007. The structure of sentence (29) is presented in (70), where the two operators making up *only* envelop the modal.

- (69) a. Morphology: *only* spells out NEG and $\exists^{exc}$
 b. Semantics: $\llbracket \exists_C^{exc} S \rrbracket = 1$ iff $\exists S' \in C \cap IE(S)$: $\llbracket S' \rrbracket = 1$
- (70) a. To get good cheese, you only have to go to the North End_F.
 b. $[exh_{C'} [_{S'} \text{NEG} [\square [\exists_C^{exc} [_{S'} \text{you go to the NE}_F]]]]]$

On the assumption that we care about three main locations, the North End, New York City and France, the domains of *exh* and $\exists^{exc}$ can be selected as in (71). This is because effort minimization in the interpretation of teleological modals makes having to go to the North End be incompatible with having to go to New York City, etc; hence, pruning of all but the strongest alternative is admitted, as otherwise the Informativity Constraint would be violated.

- (71) Domain resolution in (70b):
 a. $C \cap F(S) = \{[\text{you go to the NE}_F], [\text{you go to NYC}_F], [\text{you go to France}_F]\}$
 b. $C' \cap IE(S') = \{[\text{NEG} [\square [\exists_C^{exc} [\text{you go to Japan}_F]]]]\}$

The resulting meaning of (70b) is computed in (72). The positive presupposition corresponds to the negation of the excludable alternative in (71b), that is, you have to go to the North End, New York City, or France. The assertive meaning of the sentence is that you do not have to go to New York City or France. This corresponds to the target sufficiency reading of the sentence – to get good cheese, it suffices to go to the North End (see von Fintel & Iatridou 2005, 2007, von Fintel 2024, and references therein, for an extensive discussion of the notion of sufficiency).

- (72) Assertion of (70b) on the resolution in (71):

$$\llbracket [\text{NEG} [\square [\exists_C^{exc} [\text{you go to the NE}_F]]]] \rrbracket = 1$$

$$\Leftrightarrow \neg \square (\exists S \in \{ [\text{you go to NYC}], [\text{you go to France}] \} : \llbracket S \rrbracket = 1)$$

$$\Leftrightarrow \neg \square (\text{you go to NYC} \vee \text{you go to France})$$

- (73) Presupposition of (70b) on the resolution in (71):

$$\neg \exists S \in C' \cap IE(S') : \llbracket S \rrbracket = 1$$

$$\Leftrightarrow \neg \llbracket [\text{NEG} [\square [\exists_C^{exc} [\text{you go to the Japan}_F]]]] \rrbracket$$

$$\Leftrightarrow \square (\text{you go to the NE} \vee \text{you go to NYC} \vee \text{you go to France})$$

Necessity readings

The sentences that have the surface form of the sufficiency modal constructions do not always convey a sufficiency reading – they often only convey the stronger, necessity reading. This is demonstrated by sentence (74), which can only describe the requirement to read *War and Peace* in the literature class. (The reading is even more pronounced with some other modals, like *be required to*, as discussed by von Fintel & Iatridou 2007.)

- (74) In the literature class, we only have to read *War and Peace*.
 $\Rightarrow$ In the literature class, we have to read *War and Peace*

This stronger reading can be derived on the distributed analysis from two LFs: (i) by having $\exists^{exc}$ scope above the modal, or (ii) by having it scope below the modal. In both cases, pruning is not admitted as the sentence satisfies the Informativity Constraint without pruning (one may be required to read more than one book for a class). We focus on the latter derivation in the following, as the former derivation parallels those discussed above. On this derivation, the sentence has the structure in (75).

- (75) $[\text{exh}_{C'} [{}_S \text{ NEG } [\Box [\exists_C^{exc} [{}_S \text{ we read War and Peace}_F]]]]]$

Given that the effort minimization does not feature in the interpretation of the modal, the alternatives to the sister of *exh* are contextually mutually compatible, and the Informativity Constraint is satisfied. This means that no pruning is admitted, so the domain of *exh* is of the form provided in (76), for example.

- (76) Domain resolution for (75):
 a. $C \cap F(S) = \{[\text{we read } X_F] \mid X \in F(WP_F)\}$
 b. $\{[\text{NEG } [\Box [\exists_C^{exc} [\text{we read } X_F]]]] \mid X \in F(WP_F) \wedge [[X]] \otimes WP\} \subseteq C' \cap IE(S')$

On this domain resolution, the structure in (75) has the interpretation in (77)-(78). First, the sentence asserts that we do not have to read a book distinct from *War and Peace*. Second, the sentence presupposes that for every plurality of books not overlapping with *War and Peace*, we must read a book that does not overlap with it. This means that for every set of books that includes *War and Peace*, including the singleton set, we must read a book in that set. Accordingly, the presupposition is equivalent to the requirement that we read *War and Peace*.

- (77) Assertion of (75) on the resolution in (76):

$$\neg \Box (\exists S' \in C \cap IE(S): [[S']] = 1 \\ \Leftrightarrow \neg \Box (\exists x: x \otimes WP \wedge \text{we read } x)$$

- (78) Presupposition of (75) on the resolution in (76):

$$\forall x: x \otimes WP \rightarrow \Box (\exists y: y \otimes x \wedge \text{we read } y) \\ \Leftrightarrow \Box (\text{we read } WP)$$

This concludes our discussion of sufficiency modal constructions. We have focused exclusively on the positive inference in these cases, remaining within the framework of von Fintel & Iatridou 2007 except for updating their analysis of *only* to a distributed one. Other puzzling aspects of the construction remain, all of which cut across the central issue of this paper – or so we believe. For more comprehensive treatments, see von Fintel & Iatridou 2007, Enguehard 2021, Alonso-Ovalle & Hirsch 2022, and the references therein.

3.5 Excursus: Weak positive presupposition more broadly?

Our proposal has derived Klinedinst's generalization about the distribution of weak positive presuppositions, repeated below: weak positive presuppositions can only be derived if the excludable alternatives are mutually contextually incompatible.

(30) **Weak and Strong Content:**

An *only* sentence induces a weak positive presupposition if the focus alternatives to the sister of *only* are mutually incompatible in the context. Otherwise the sentence induces a strong positive inference, which corresponds to the prejacent of *only*.

The generalization may, however, appear too strong. A reviewer draws attention to examples such as (79), noting that they too seem to trigger merely weak positive presuppositions, at least for some speakers. The reviewer correctly observes about the example that the relevant actions under consideration are, in principle, mutually compatible – for instance, one could both sell hallucinogenic mushrooms and sell champignons.

(79) [Context: Jack was arrested for selling hallucinogenic mushrooms, which is against the law. I complain that this is very unfair. My interlocutor responds:]

Well, Jack did not only sell champignons_F.

Observed positive inference: Jack sold some kind of mushrooms
(not: Jack sold champignons)

We contend that both the potential availability of a weak positive presupposition in examples such as (79), and its somewhat elusive nature, can be captured by the proposal developed above. In particular, we put forward that to the extent that one gets a weak positive presupposition in (79), selling champignons is taken to pick out an activity, or a class of activities, that crucially does not include selling hallucinogenic mushrooms. Since the two activities are in fact compatible, such a disjoint classification can be derived via embedded exhaustification. Accordingly, the proposed parse of the sentence in (79) is as given in (80). If only the alternatives concerning the selling of hallucinogenic mushrooms are included in *C*, the sister of *only* in (80) receives the interpretation shown in (81). (As before, because *exh* is presuppositional, the exhaustive inference must be locally accommodated in order to take effect. The more complex alternative involving hallucinogenic mushrooms is available due to its explicit mention in the preceding discourse; cf. Katzir 2007.)

(80) [neg [exh_{C''} [*S''* only_{C'} [*S'* ACCOM [exh_C [*S* Jack sold champignons_F]]]]]]

(81) a. $C \cap F(S) = \{[\text{Jack sold hallucinogenic mushrooms}]_F\}$

b. $\llbracket S' \rrbracket = 1$ iff (Jack sold champignons) $\wedge \neg$ (Jack sold hallucinogenic mushrooms)

Since the alternatives that *only* ranges over are on the above assumptions mutually incompatible, matrix exhaustification is bound to violate the Informativity Constraint along the lines described in the preceding. Accordingly, pruning of the kind provided in (82) is admitted.

(82) Domain resolution in (80):

a. $C' \cap F(S') = \{[\text{ACCOM [exh}_C \text{ [Jack sold champignons}_F \text{]}]}],$
 $[\text{ACCOM [exh}_C \text{ [Jack sold [haluc. mushrooms]}_F \text{]}]]\}$

b. $C'' \cap \text{IE}(S'') = \{[\text{only}_{C'} [\text{ACCOM [exh}_C \text{ [Jack sold potatoes}_F \text{]}]]]\}$

The assertive meaning of the structure in (80) is given in (83): Jack sold hallucinogenic mushrooms. Its positive presupposition is weak as a result of pruning described in (82) – namely, it reduces to the claim that Jack sold some kind of mushrooms, as shown in (84). Since this presupposition does not entail the assertion, the Informativity Condition is satisfied.

(83) Assertion of (80) on the resolution in (82):

$$\begin{aligned} & \exists S \in C' \cap IE(S'): \llbracket S \rrbracket = 1 \\ & \Leftrightarrow \text{Jack sold hallucinogenic mushrooms} \end{aligned}$$

(84) Presupposition of (80) on the resolution in (83):

$$\begin{aligned} & \neg \exists S \in C'' \cap IE(S''): \llbracket S \rrbracket = 1 \\ & \Leftrightarrow \neg \llbracket [\text{only}_{C'} [S' \text{ ACCOMM} [\text{exh}_C [S \text{ Jack sold potatoes}_F]]]] \rrbracket \\ & \Leftrightarrow (\text{Jack sold champignons} \vee \text{Jack sold hallucinogenic mushrooms}) \end{aligned}$$

We have thus shown that it is, in principle, possible to derive weak positive presuppositions even in cases where the focus alternatives to the sister of *only* do not appear to be mutually contextually incompatible. We argued, however, that this appearance is misleading and that embedded exhaustification is at play. Why are such readings less readily accessible in examples like the mushroom example in (79), and many others, than in our earlier examples? There are at least two interrelated reasons for this. First, deriving the weak presupposition in examples like (79) requires inserting two additional silent operators into the structure, rendering the reading less accessible. For a hearer to converge on such a parse, then, there must be sufficient contextual cues signaling that a more complex structure is warranted. Second, a weak positive presupposition is accompanied by a scalar inference, which we have set aside for the most part, according to which the prejacent is ranked lower than other alternatives on some salient scale. While such an inference may be accommodated in the mushroom-selling example, where selling only champignons is ranked below selling hallucinogenic mushrooms on an unlawfulness scale, accommodation of an appropriate scale is not possible in many other cases where embedded exhaustification could, in principle, facilitate a weak presupposition.

This wraps up the derivation of the weak positive presuppositions on the distributed approach of *only*. We argued that their availability follows from a single feature of the analysis – namely, the ability to constrain what alternatives participate in the derivation of the positive inference. In addition, the restricted distribution of weak positive presuppositions was shown to be derivable by appealing to natural, independently motivated constraints on the pruning of alternatives.

4 Integrated approaches

Weak positive presuppositions are, all else being equal, unexpected on the integrated analysis of *only* presented in Sect. 1. Two broad strategies have been pursued to account for them within that framework. The first strategy retains Horn’s prejacent presupposition analysis outlined above and mitigates the strength of the positive presupposition, either through local accommodation or by invoking a special attenuation operator. We discuss these approaches in Sect. 4.1. The second strategy takes the opposite tack: it weakens Horn’s analysis and derives stronger readings either by means of exhaustification or by adopting special assumptions about the alternatives quantified over by *only*. We discuss these approaches in Sect. 4.2. As will become clear, the approaches share certain points of contact with the distributed analysis (cf. multiple operators, domain restriction), though they also diverge from it in important ways. The supplementary assumptions they have to

invoke are nontrivial and, we argue, their nature lends support to the distributed analysis. It goes without saying, however, that none of the approaches can be done full justice here.

4.1 Strong to weak

If we retain the lexical entry for *only* in (2), repeated below, we generate the wrong inferences for sentences (25)-(28); specifically, we generate inferences that are contextual contradictions.

(2) An integrated analysis of *only*:

$\llbracket \text{only}_C S \rrbracket$ is defined only if $\llbracket S \rrbracket = 1$.

If defined, $\llbracket \text{only}_C S \rrbracket = 1$ iff $\neg \exists S' \in C \cap \text{IE}(S): \llbracket S' \rrbracket = 1$.

For example, sentence *Gal isn't only a grad student_F* in (25) is incorrectly predicted to presuppose that Gal is a grad student (rather than that she is a grad student, a postdoc, or a professor) and to assert that she is either a postdoc or a professor, as given in (86) and (87). Together, these inferences yield a contextual contradiction.

- (85) a. Gal isn't only a grad student_F.
b. $\llbracket \text{not} [\text{only}_C [\text{Gal is a grad student}]_F] \rrbracket$

- (86) Assertion of (85b):
(Gal is a postdoc $\vee$ Gal is a professor)

- (87) Presupposition of (85b):
Gal is a grad student

In response, we can either (i) suspend the strong positive presupposition and derive the weak positive presupposition in some other way, or (ii) keep the strong positive presupposition and weaken the prejacent of *only*. We discuss these possibilities in turn.

(i) Accommodation

One strategy for addressing the challenge of weak positive presuppositions is to assume, first, that the strong positive presupposition is locally accommodated within the scope of negation or the question operator in examples such as (25)-(28). This explains why the strong positive presupposition does not project out of these sentences, avoiding a contextual contradiction. At the same time, since the sentences do give rise to an existential positive presupposition, this inference must be derived by other means. We can achieve this, second, by taking the inference to be generated by focus (cf. Geurts & van der Sandt 2004, Abusch 2010). Accordingly, the sentence in (25) can be assigned the LF in (88): the accommodation operator ACCOMM locally accommodates the strong positive presupposition, while FOCUS triggers the presupposition that some focus alternative to its sister is true.

- (88) $\llbracket \text{neg} [\text{FOCUS}_{C'} [S' \text{ ACCOMM} [\text{only}_C [S \text{ Gal is a grad student}]_F]] \rrbracket$

On these assumptions, the LF in (88) triggers the focus presupposition that Gal is at least a grad student, which projects under negation, as provided in (89). The asserted meaning is then the negation of Gal being a grad student and nothing else, which is equivalent to either Gal not being a grad student or her being a postdoc or a professor, as provided in (90). Together with the focus presupposition, this meaning is equivalent to Gal being either a postdoc or a professor.

(89) Presupposition of (88) (due to FOCUS):

$$\exists S \in C' \cap F(S'): \llbracket S \rrbracket = 1$$

$$\Leftrightarrow (\text{Gal is a grad student} \vee \text{Gal is a postdoc} \vee \text{Gal is a professor})$$

(90) Assertion of (88):

$$\neg \llbracket \text{ACCOMM} [\text{only}_C [\text{Gal is a [grad student]}_F]] \rrbracket$$

$$\Leftrightarrow \neg(\text{Gal is a grad student}) \vee \exists S' \in C \cap \text{IE}(S): \llbracket S' \rrbracket = 1$$

$$\Rightarrow \neg(\text{Gal is a grad student}) \vee (\text{Gal is a postdoc} \vee \text{Gal is a professor})$$

(91) Presupposition of (88) $\wedge$ Assertion of (88) $\Leftrightarrow$ (Gal is a postdoc $\vee$ Gal is a professor)

The accommodation approach runs into at least two problems. The first one is that it offers no traction with sufficiency modal constructions, which do not provide environments in which the accommodation operator could be effective (say, they do not contain negation). The second problem is more serious: the proposed account is still at odds with the Informativity Constraint, a constraint that cannot be circumvented by an insertion of ACCOM, either at the matrix or embedded level (as argued forcefully by Doron & Wehbe 2022, Guerrini & Wehbe 2025).

(42) **Informativity Constraint:**

No LF may have an assertive meaning contextually entailed by its presupposition.

Guerrini & Wehbe 2025 illustrate the application of the constraint in global contexts with sequences like (92): the common ground together with the presupposition of the second sentence, that John has 10 children, entails the second sentence, rendering it infelicitous. As potential insertion of ACCOM is ineffective here, it should therefore be ineffective in our cases as well.

(92) I knew that all of John's kids are adopted but today I discovered something amazing.

#All 10 of John's kids are adopted!

(ii) Attenuation

Alonso-Ovalle & Hirsch 2022 analyze sufficiency modal constructions by introducing a special covert attenuating operator, AT LEAST. Their primary motivation is a criticism of von Fintel & Iatridou's 2007 analysis: it attributes to *only* a weak positive presupposition, which Alonso-Ovalle & Hirsch argue is empirically inadequate. This criticism, however, is neutralized once von Fintel & Iatridou's 2007 analysis is embedded within the distributed system, as we have done above. Namely, on the distributed analysis, *only*-sentences give rise to strong positive presuppositions in default cases. Examining Alonso-Ovalle & Hirsch's proposal is nonetheless instructive.

In response to an earlier version of this paper, Alonso-Ovalle & Hirsch 2023 employ AT LEAST to weaken the positive inferences in the examples under discussion. This operator requires the meaning of its sister to be ranked lowest among the relevant alternatives, and it forms a disjunction of that meaning with the more highly ranked alternatives. For instance, (26) can be parsed as in (93), with AT LEAST occurring in the immediate scope of *only*. The meaning of the AT LEAST-prefixed phrase – which constitutes the presupposition of the sentence triggered by *only* – is given in (94). It conveys that Gal being a grad student is ranked lower than the alternatives, (94a), and that Gal is a grad student, a postdoc, or a professor, (94b).

(93) [not [only_{C'} [AT LEAST_C [Gal is a [grad student]_F]]]

(94) Presuppositions of (93):

- a. $\forall S \in C \cap IE([Gal \text{ is a } [grad \text{ student}]_F]):$
 $[Gal \text{ is a } [grad \text{ student}]_F] <_c S$
- b. $[[[AT \text{ LEAST}_C [Gal \text{ is a } [grad \text{ student}]_F]]] = 1$
 $\Leftrightarrow \exists S \in C \cap F([Gal \text{ is a } [grad \text{ student}]_F]): [[S]] = 1$
 $\Leftrightarrow (Gal \text{ is a grad student} \vee Gal \text{ is a postdoc} \vee Gal \text{ is a professor})$

In (93), the relevant excludable alternatives for *only* are that Gal is at least a postdoc and that she is a professor (we must set aside the presuppositions that *AT LEAST* triggers in the alternatives, as they contradict the presupposition of the sentence). These alternatives are negated by *only*, and the result is subsequently negated by matrix negation. This yields the assertive meaning in (95): that Gal is at least a postdoc. The proposal thus derives both the weak positive presupposition and the target assertive meaning.

(95) Assertion of (93):

$$\neg(\neg\exists S \in C' \cap IE([AT \text{ LEAST}_C [Gal \text{ is a } [grad \text{ student}]_F])): [[S]] = 1)$$

$$\Leftrightarrow (Gal \text{ is a postdoc} \vee Gal \text{ is a professor})$$

The attenuation approach faces problems of overgeneration, as noted already by Alonso-Ovalle & Hirsch 2022, 2023. First: *AT LEAST* is unexpectedly absent in environments without *only*, that is, the approach posits an operator whose distribution is tied specifically to *only* and the inferences under discussion; it is, hence, unlike the operators adopted on either the accommodation or the distributed analysis. This also sets *AT LEAST* apart from von Stechow & Iatridou's covert $\exists^{exc}$, which is not optional and has been argued to be overtly realized in exclusive expressions in some languages (e.g., French, Greek, Irish, Romanian). Although Alonso-Ovalle & Hirsch do partly defend *AT LEAST* by appealing to expressions in other languages (cf. Schwarz 2005, Crnič 2011), these other expressions are scalar additives – that is, they do not occur with exclusives, and so do not obviously offer cross-linguistic support for an *AT LEAST* operator tied to *only*.

Second: The attenuation approach predicts weaker meanings than observed even in the presence of *only*. For instance, the parse in (96b) incorrectly predicts that *Gal didn't only read War and Peace_F* has a reading that fails to entail that she read War and Peace.

- (96) a. Gal didn't only read War and Peace_F.
- b. $[neg [only_{C'} [AT \text{ LEAST}_C [Gal \text{ read War and Peace}_F]]]]$

Accordingly, Alonso-Ovalle & Hirsch 2023 propose that *AT LEAST* can be inserted only to resolve pragmatic anomalies, such as violations of the Informativity Constraint. This condition still does not adequately curb overgeneration, however. Consider the infelicitous (97), which triggers contextually inconsistent presuppositions, namely, that Gal is simultaneously a grad student and a postdoc or a professor.

(97) #Tal only knows that Gal is a grad student.

This infelicity is not expected on the attenuation approach, all else equal. Consider the following scenario: Gal is a professor; Tal justifiably believes that Gal is at least a grad student; and she correctly believes that the three statuses (grad student, postdoc, professor) are mutually exclusive. that you can only be one of grad student, postdoc, or professor. In such a context, the attenuation approach plus the pragmatic anomaly constraint predicts that the parse in (98) should

be acceptable: the insertion of AT LEAST averts a pragmatic anomaly, the resulting inferences are all consistent, and the presuppositions of all the alternatives in the domain of *only* are satisfied in the context.

(98) [only_{C'} [Tal knows [AT LEAST_C [Gal is a [grad student]_F]]]]

(99) Presupposition of (98):

Tal knows Gal is a grad student or a postdoc or a professor

(100) Assertion of (99b):

$\neg(\text{Tal knows Gal is a postdoc or a professor}) \wedge \neg(\text{Tal knows Gal is a professor})$

This calls for an additional constraint on the structural relation between *only* and AT LEAST. However, the constraint cannot simply require *only* and AT LEAST to be adjacent at LF – as they are in (93) but not in (98) – since their scope must be split to derive the correct readings of sufficiency modal constructions. This is illustrated in (101) (see Alonso-Ovalle & Hirsch 2022 for details): the requirement is that you go at least to the North End, that you need not go anywhere else, and that going to the North End is the lowest-ranked option among the alternatives.

- (101) a. To get good cheese, you only have to go to the North End.
 b. [only_{C'} [have [AT LEAST_C [you go to the North End]]]]

While we cannot attempt to formulate an adequate constraint here, a comparison with our rendition of von Fintel & Iatridou, which faces the same issue, is instructive regarding to the difficulty of this task. Recall that on that approach, the operator that may be assigned low scope relative to the factive verb is the exceptive $\exists^{exc}$, while negation and *exh* take matrix scope, as given in (102).

(102) [exh_{C'} [NEG [Tal knows [$\exists^{exc}_C$ Gal is a grad student]_F]]]]

We focus on two points of comparison. First, our rendition of the von Fintel & Iatridou approach can point to a suggestive parallelism: *only*-counterparts in languages like French and Greek, which are made up of negation and an overt exceptive operator, share this very restriction – a factive verb cannot split the negation and the exceptive (the exceptive independently exhibits the distribution of an NPI, as discussed by von Fintel & Iatridou 2007; cf. Chierchia 2013, Ch. 7, on factive intervention). Since these overt counterparts of $\exists^{exc}$ are uncontroversially exceptive, an analysis of them as AT LEAST is not tenable (for example, a consequence of such an analysis would be that weak positive presuppositions occur across the board, contrary to fact). The second point builds on the first. Our rendition of the von Fintel & Iatridou approach allows for a characterization of the unavailability of the split in (102) that aligns naturally with independent work on markedness (esp., Magri 2009, 2011, Anvari 2018, Bar-Lev 2024). Specifically, exhaustification over the full set of formal alternatives to the sister of *exh* in (102) yields a contextual contradiction, as the presuppositions of the excludable alternatives are mutually contextually incompatible (this is possible because excludability is determined on the basis of logical rather than contextual compatibility) – and such structures are marked elsewhere. Although a full explanation of the limits on von Fintel & Iatridou's $\exists^{exc}$ remains beyond the scope of this paper, we note that referencing formal alternatives within the attenuation approach, whether for *only* or AT LEAST, is unhelpful: quantifying over the full set of formal alternatives there yields a consistent meaning, as we have seen. Accordingly, it is less clear what explanatory generalization the attenuation approach might

offer beyond mere description. (See Condoravdi & Francez 2022, as well as Sect. 5 below, for discussion of additional issues.)

Summary

The integrated analysis of *only* on which *only* triggers a strong positive presupposition fails to adequately account for the distribution of the positive inferences of *only*. When supplemented with the accommodation approach, the analysis undergenerates (e.g., in sufficiency modal constructions) and fails to explain how certain acceptable *only* sentences comply with the Informativity Constraint. The attenuation approach offers an advance on these issues, but it itself overgenerates. While constraints could be formulated to rein in this overgeneration, these would be tailored exclusively to the *only* sentences under discussion, raising questions about the explanatory reach of the proposal. Finally, it bears re-emphasizing that transposing von Fintel & Iatridou’s 2007 analysis into the distributed system neutralizes the central objection raised against it by the attenuation approach – namely, that it assigns *only* a positive presupposition that is too weak.

We turn now to an alternative integrated analysis of *only*.

4.2 Weak to strong

Only has also been analyzed as triggering a weak positive presupposition (e.g., Horn 1996, von Fintel 1997, Klinedinst 2005, Beaver & Clark 2008, among others). Such a definition is given in (103): the exclusive inference is as above, while the positive presupposition involves existential quantification over relevant focus alternatives (cf. Horn 1996, von Fintel 1997).

(103) An integrated analysis of *only* (weak):

$\llbracket \text{only}_C S \rrbracket$ is defined only if $\exists S' \in C \cap F(S)$: $\llbracket S' \rrbracket = 1$.

If defined, $\llbracket \text{only}_C S \rrbracket = 1$ iff $\neg \exists S' \in C \cap IE(S)$: $\llbracket S' \rrbracket = 1$.

The proposal accounts for simple unembedded occurrences of *only*. For example, sentence (1), repeated below, is predicted to have the meaning in (104)-(105): it presupposes that someone was at the party, and asserts that no one distinct from Gal was at the party. Together, these inferences entail the strong positive inference – that Gal was at the party.

(1) Only Gal_F was at the party.

Positive inference: Gal was at the party

Exhaustive inference: No one distinct from Gal was at the party

(104) Presupposition of (1):

$\exists S \in C \cap F(\text{Gal}_F \text{ was at the party})$: $\llbracket S \rrbracket = 1$

$\Leftrightarrow \exists x$: x was at the party

(105) Assertion of (1):

$\neg \exists S \in C \cap IE([\text{Gal}_F \text{ was at the party}])$: $\llbracket S \rrbracket = 1$

$\Leftrightarrow \neg \exists x$: $x \otimes \text{Gal} \wedge x$ was at the party

(106) Presupposition of (1) $\wedge$ Assertion of (1) $\Rightarrow$ Gal was at the party

The proposal further accounts for the projection facts in (25)-(28), where the sentences convey merely a weak positive inference. For example, sentence *Gal isn't only a grad student_F* is predicted to have the meaning in (108)-(109): it presupposes that Gal is either a grad student, a postdoc, or a professor, and asserts that she is either a postdoc or a professor.

- (107) a. Gal isn't only a grad student_F
 b. [neg [only_C [Gal is a [grad student]_F]]]
- (108) Presupposition of (107b):
 (Gal is a grad student $\vee$ Gal is a postdoc $\vee$ Gal is a professor)
- (109) Assertion of (107b):
 (Gal is a postdoc $\vee$ Gal is a professor)

The main problem for the analysis stems from embedded occurrences of *only* that generate strong positive presuppositions (but see Alonso-Ovalle & Hirsch 2022 for an extensive discussion of other problems). For example, sentence *Not only Gal_F was at the party* is predicted to presuppose merely that someone was at the party, as shown in (111). This is weaker than the observed positive inference of the sentence – namely, that Gal was at the party.

- (110) a. Not only Gal_F was at the party.
 b. [neg [only_C [Gal_F was at the party]]]
- (111) Presupposition of (110b):
 $\exists x: x$ was at the party
- (112) Assertion of (110b):
 $\exists x: x \otimes \text{Gal} \wedge x$ was at the party
- (113) Presupposition of (110b) $\wedge$ Assertion of (110b) $\nRightarrow$ Gal was at the party

A separate account is accordingly needed for such sentences. We discuss two approaches: (i) strengthening the meanings generated by *only* via exhaustification, and (ii) further revising the definition of *only* so that its domain is appropriately restricted to yield stronger positive presuppositions in certain cases.

(i) Exhaustification

The stronger positive presupposition can be derived by means of exhaustification. Sentence *Not only Gal_F was at the party* could be assigned the structure in (114), and stipulated to have the sentence without *only* as an alternative, but not the sentence without *not only* (as having both would make them both not excludable). Accordingly, this *only*-less alternative would be excludable, and the strong positive presupposition would be derived, as provided in (114).

- (114) [exh_{C'} [_S neg [only_C [Gal_F was at the party]]]]
- (115) If it holds that: [neg [Gal was at the party]] $\in C' \cap \text{IE}(S)$, then
 [[exh_{C'} [neg [only_C [Gal_F was at the party]]]] is defined only if
 [[neg [Gal_F was at the party]]] = 0, that is, only if Gal was at the party

The exhaustification approach faces several problems. First, exhaustification has to apply obligatorily in all (and only) examples in which the alternatives to the sister of *exh* are mutually compatible, even if this contradicts the information in the common ground. Such obligatoriness is at odds with what is usually assumed about exhaustification, though it is in line with the distributed analysis of *only*. Second, and more importantly, the approach does not generalize beyond embedding under negation. For example, consider the conditional sentence in (116), in which *only* occurs in the antecedent of a conditional sentence. The sentence may generate a strong positive presupposition, that Gal was at the party (the positive presupposition can also be accommodated in the antecedent; cf. Tonhauser et al. 2013, Crnič 2024 for related discussion).

(116) If only Gal_F was at the party, the party was a failure.

Possible strong positive presupposition: Gal was at the party

If we attempt to exhaustify the sentence at the matrix level in (116), one negates a complete conditional sentence, the counterpart of (116) without *only*. The resulting inference (= it is false that if Gal was at the party, the party was a failure) not only fails to license an inference to Gal having been at the party, it is clearly not an inference that the sentence gives rise to at all. What about exhaustifying at the embedded level? One would have to conjure a different set of alternatives to derive the desired result. For example, those assumed on the distributed analysis would work. But this would simply reduce the proposal to the distributed analysis of *only*, but with the extra assumption that *only* lexically encodes a weak positive presupposition.

(ii) Domain restriction via pragmatic scales

Rather than analyzing the positive presupposition as involving simple existential quantification over focus alternatives, as in (103), one could instead restrict it to a specific subset of focus alternatives – say, those ranked in a specific way (esp., Jacobs 1983, Krifka 1993, Klinedinst 2005, Beaver & Clark 2008, Coppock & Beaver 2014). One additional motivation for introducing such a restriction is the tendency of *only* to give rise to a scalar inference, namely, that its prejacent occupies a low position on a salient scale (see Sect. 2 above, and Greenberg 2022 for a recent review). On this approach, the weak positive inference and the scalar inference are closely connected.

A schematic representation of the revision is in (117). The additional constraint on alternatives is represented by a Scale Condition function, underlined in (117), which relates the sister of *only* to the set of relevant alternatives. The task is now to calibrate this function appropriately.

(117) **An integrated analysis of *only* (weak + scale condition):**

$\llbracket \text{only}_C S \rrbracket$ is defined only if $\exists S' \in C \cap F(S)$: $\llbracket S' \rrbracket = 1$ and Condition(S)(C) = 1.

If defined, $\llbracket \text{only}_C S \rrbracket = 1$ iff $\neg \exists S' \in C \cap IE(S)$: $\llbracket S' \rrbracket = 1$.

Case-based Scale Condition. The meaning of *only* in (117) can capture the complex behavior of *only*'s positive inference, provided the Scale Condition is appropriately specified. For sentences like (1), which generate a strong positive presupposition, the meaning in (117) is adequate if all relevant focus alternatives over which existential quantification ranges entail the sister of *only* (esp., Klinedinst 2005, Beaver & Clark 2008). The Scale Condition function can encode this, as illustrated in (118): only the alternatives that entail the sister of *only* are considered in computing the positive inference – that is, *only* selects for a resource domain whose elements form an entailment scale. (Existential quantification over at-least-as-highly-ranked alternatives is conventionally represented with MIN; the corresponding exclusive inference also refers to ranked alternatives and

is represented with MAX. The latter, however, is not directly relevant here. See, e.g., Beaver & Clark 2008, Ch. 10, and Coppock & Beaver 2014 for definitions and application.)

(118) **Scale Condition (to be revised):**

$$\text{Condition}(S)(C) = 1 \text{ iff } \forall S' \in C \cap F(S): \llbracket \wedge S' \rrbracket \Rightarrow \llbracket \wedge S \rrbracket$$

One set of alternatives that satisfies the Scale Condition for sentence (1), *Only Gal_F was at the party*, together with the corresponding presupposition of the sentence, is provided in (119): it consists of the alternatives in which *Gal* is replaced with expressions denoting individuals that include Gal. As desired, the presupposition derived on this selection of alternatives is equivalent to Gal having been at the party – the strong positive inference. Namely, each alternative that the existential quantification ranges over in (119) entails that Gal was at the party; hence, one of them being true holds iff Gal was at the party (which corresponds to the weakest of the alternatives).

(119) If $C \cap F(S) = \{[X \text{ was at the party}] \mid \text{Gal} \sqsubseteq \llbracket X \rrbracket\}$, then

$\llbracket [\text{only}_C [S \text{ Gal}_F \text{ was at the party}]] \rrbracket$ is defined only if

$$\exists S \in C \cap F([\text{Gal}_F \text{ was at the party}]): \llbracket S \rrbracket = 1$$

$$\Leftrightarrow \text{Gal was at the party}$$

The Scale Condition must be weakened, however, to admit weak positive inferences in the cases discussed above. At the same time, it cannot be weakened too much, since the strong positive inference in (119) must still be preserved. One intuitive formulation of such a weakening is that the scale of alternatives is resolved to an entailment scale – logically stronger alternatives are ranked at least as high as logically weaker ones – unless it cannot be (e.g., Klinedinst 2005, Beaver & Clark 2008, Coppock & Beaver 2014). A proper formulation of this condition is not a trivial matter. We rely on the rough approximation in (120), despite its shortcomings. (For example, strictly speaking, *grad student* may have a stronger alternative *MA student*, which should then force an entailment-based scale resolution. And further issues arise when focused expressions only have logically independent alternatives, which would incorrectly be predicted to result in weak positive presuppositions. To our knowledge, no fully adequate formulation of the Scale Condition has been proposed in the literature.)

(120) **Scale Condition (final approximation):**

$$\text{Condition}(S)(C) = 1 \text{ iff } [\forall S' \in C \cap F(S): \llbracket \wedge S' \rrbracket \Rightarrow \llbracket \wedge S \rrbracket] \vee \quad (\text{entailment ranking ...})$$

$$[(\forall S' \in F(S): S \neq S' \rightarrow \llbracket \wedge S' \rrbracket \not\Rightarrow \llbracket \wedge S \rrbracket) \wedge \quad (\dots \text{ unless impossible})]$$

$$(\forall S' \in C \cap F(S): \llbracket \wedge S \rrbracket \leq_c \llbracket \wedge S' \rrbracket)]$$

The strong inference in (119) remains in place on this revision – namely, one selects focus alternatives that are ranked by entailment if this is possible. But now *only* sentences with weak positive inferences are admitted in addition. For example, the sentence in (26), *Gal is not only a grad student_F*, now triggers the presupposition provided in (121), where the pertinent scale is based on the status of the different stages of the usual academic path. The presupposition corresponds to the observed weak positive inference of the sentence.

(121) If $C \cap F(S) = \{[\text{Gal is a [grad student]}_F], [\text{Gal is a postdoc}_F], [\text{Gal is a professor}_F] \}$,

$\llbracket [\text{only}_C [S \text{ Gal is a [grad student]}_F]] \rrbracket$ is defined only if

$$\exists S \in C \cap F([\text{Gal is a [grad student]}_F]): \llbracket S \rrbracket = 1 \wedge \llbracket \wedge \text{Gal is a grad student} \rrbracket \leq_c \llbracket \wedge S \rrbracket$$

$$\Leftrightarrow \text{Gal is at least a grad student}$$

To sum up, the proposal accounts for the distribution of the positive inferences of *only* as follows: when a domain restriction of focus alternatives that entail the prejacent can be accommodated, it must be – giving rise to a strong positive presupposition. When such a restriction cannot be accommodated, the alternatives must instead be ranked relative to a non-entailment scale – giving rise to a weak positive presupposition.

The challenge. The analysis encounters a difficulty with sentences in which the associate of *only* does not occur in its immediate scope. To see this, consider the sentence in (122), where *only* takes matrix scope while the focused expression appears in the antecedent of a conditional. This sentence gives rise to a strong positive presupposition, as shown by its negated and interrogative variants in (123), both of which entail the strong positive inference.

(122) Only if Gal_F was at the party was the party a success.

Observed positive inference: If Gal was at the party, the party was a success

(123) a. Not only if Gal_F was at the party was the party a success.
b. Was the party a success only if Gal_F was at the party?

Observed positive inference: If Gal was at the party, the party was a success

The focus alternatives to the sister of *only* in (123) are provided in (124), where the focused proper name is replaced by other DPs, say, *if Tal was at the party, the party was a success*, or *if the girls were at the party, the party was a success*.

(124) $F(\text{[if Gal}_F \text{ was at the party, the party was a success]}) =$
 $\{\text{[if X was at the party, the party was a success]} \mid X \in F(\text{Gal}_F)\}$

Importantly, no focus alternative to the sister of *only* distinct from it entails it, which holds no matter what semantics of conditionals one assumes, as stated in (125): for example, the proposition that if Gal and Tal were at the party, the party was a success is either entailed by the sister of *only* (under a strict analysis of conditionals, cf. von Stechow 2001, Gillies 2007) or independent from it (under a variably strict analysis of conditionals, cf. Stalnaker 1968, Lewis 1973, Kratzer 1986) – what is crucial is that it does not entail that if Gal was at the party, the party was a success.

(125) For all $S \in F(\text{[if Gal}_F \text{ was at the party, the party was a success]})$:
 $\text{[if Gal}_F \text{ was at the party, the party was a success]} \neq S \rightarrow$
 $\llbracket \wedge S \rrbracket \not\Rightarrow \llbracket \wedge \text{[if Gal}_F \text{ was at the party was the party a success]} \rrbracket$

The sentence in (122) is thus predicted to trigger the presupposition that at least one alternative ranked as high as the sister of *only* is true, as in (126). Since no focus alternative distinct from the sister of *only* entails it, the ranking obviously cannot be entailment-based. As a result, the presupposition leaves open whether it holds that if Gal was at the party, the party was a success.

(126) Predicted presupposition of (122):
 $\exists S \in C \cap F(\text{[if Gal}_F \text{ was at the party, the party was a success]}) : \llbracket S \rrbracket = 1 \wedge$
 $\llbracket \wedge \text{[if Gal}_F \text{ was at the party, the party was a success]} \rrbracket \leq_c \llbracket \wedge S \rrbracket$
 $\Rightarrow (\text{if Gal was at the party, the party was a success})$

This prediction is incorrect: as demonstrated in (123), embedding the sentence under negation and in questions reveals that it triggers a strong positive presupposition – that if Gal was

at the party, the party was a success. For principled reasons, then, the approach faces the same difficulty as the initial weak-positive-presupposition account in (103) when applied to more complex sentences such as (122) – that is, sentences in which *only* is separated from its associate by intervening operators pose a substantive challenge to the domain restriction analysis.

Potential patch? One possible way to address the challenge would be to further revise the domain restriction of *only*, as suggested by a reviewer. Instead of assuming that *only* quantifies over a subset of focus alternatives defined in (124) (following, e.g., Horn 1972, Jackendoff 1972, Rooth 1992), one could assume that it quantifies over the conjunctive closure of this set. In the present case, this would require restricting the set of alternatives to the conjunctions of the pertinent focus alternatives with the prejacent – for example, the alternatives corresponding to if Tal/Gal and Tal/etc. were at the party, the party was a success, would have to be pruned. This restriction to alternatives consisting of the prejacent conjoined with other focus alternatives would derive the strong positive presupposition – recall that all relevant alternatives must entail the prejacent to derive the strong positive presupposition. We have now arrived at the system of assumptions summarized in (127). Although the complexity of this setup merits explanation, we will not pursue one here. The reason is that the combination of assumptions results in severe undergeneration – most notably, it rules out the generation of weak positive presuppositions altogether.

(127) **Assumptions about the domain restriction of *only* following the patch:**

- a. *Only*'s domain restriction is drawn from the conjunctive closure of the set of focus alternatives to the sister of *only* (*pace*, e.g., Horn 1972, Rooth 1985, 1992).
- b. *Only*'s domain restriction consists of propositions ranked by entailment, unless this is not possible (in which case the ranking is determined by context). See (120).

To illustrate the undergeneration, consider the sentences in (128). Suppose Gal, a gymnast, competes in five events at the Olympics but wins only one medal. While the sentence in (128a) can felicitously describe one instantiation of such a scenario, the sentence in (128b) cannot.

- (128) a. Gal only won a bronze_F medal.
b. #Gal only won a gold_F medal.

The felicity contrast in (128) intuitively arises from the fact that Gal's winning a gold medal is ranked higher than all other focus alternatives. On the initial domain restriction approach, on which *only* quantifies over a subset of focus alternatives, the infelicity of (128b) follows naturally: the presupposition (corresponding to Gal having won a gold medal, as this is the highest ranked alternative) entails the assertion, which is a contextual tautology (that Gal did not win a medal more notable than gold). Strikingly, this contrast can no longer be derived on the patched approach. In particular, in the second sentence, the alternatives quantified over by *only* now have meanings given in (129) – and each of these entails the prejacent and thus ranks at least as high as it.

(129) **Revised domain of quantification of *only*_C in (128b):**

- $$\{ \llbracket \wedge \text{Gal won a gold} \rrbracket, \llbracket \wedge \text{Gal won a gold} \rrbracket \wedge \llbracket \wedge \text{Gal won a silver} \rrbracket, \\ \llbracket \wedge \text{Gal won a gold} \rrbracket \wedge \llbracket \wedge \text{Gal won a bronze} \rrbracket, \\ \llbracket \wedge \text{Gal won a gold} \rrbracket \wedge \llbracket \wedge \text{Gal won a silver} \rrbracket \wedge \llbracket \wedge \text{Gal won a bronze} \rrbracket \}$$

The resulting presupposition and assertion, now both true in the described scenario, are provided in (130)-(131): that Gal won a gold medal is presupposed, and that she did not win any other medal is asserted. The sentence should, accordingly, be felicitous, contrary to fact.

(130) Revised presupposition of (128b):

$\exists p \in (129): p = 1$

$\Leftrightarrow$ Gal won a gold medal

(131) Assertion of (128b):

$\neg \exists p \in (129): p = 1 \wedge \wedge p \Rightarrow \llbracket \wedge \text{Gal won a gold} \rrbracket$

$\Leftrightarrow$ Gal won no other medal than gold

More generally, the introduction of the conjunctive closure patch to the analysis of the focus operator *only* has two significant and adverse consequences. First, the constraint generally prevents the derivation of non-entailment-ranked readings of *only* whenever the focus alternatives are mutually independent, as just illustrated. Second, and more critically, in cases where the alternatives are mutually contextually incompatible – that is, in cases where we in fact observe weak positive presuppositions – the revision incorrectly predicts that such sentences should be infelicitous due to a violation of the Informativity Constraint. The reason for this failure is straightforward: in these cases, all alternatives that *only* ranges over – besides the prejacent – are contextual contradictions (as they involve conjunctions of mutually contextually incompatible presuppositions), so only the prejacent is at least as highly ranked as itself and can be true. Since all other alternatives are contextual contradictions, their negation at the assertive level is a contextual tautology. Hence, the presupposition contextually entails the assertive meaning.

In short, not only does the revision undergenerate with respect to readings observed when the alternatives are mutually independent, it also undoes the derivation of weak positive presuppositions altogether. We refrain from further gerrymandering the characterization of *only*'s domain restriction, since this would at best amount to a descriptive patch. We conclude that cases in which *only* and its associate are separated by certain operators continue to pose a challenge for the domain restriction approach and require further attention there.

Summary

The integrated analyses of *only* that assume that *only* triggers a weak positive presupposition face serious issues in accounting for the distribution of the strong positive inferences of *only*. This can be appreciated by looking at occurrences of *only* in environments other than negated sentences (*pace* the exhaustification approach) or at occurrences of *only* that are separated from their focus associates by intervening operators (*pace* the domain restriction approach).

A deep question that remains to be addressed on the domain restriction approach is why the Scale Condition should require one to resolve the domain of *only* to alternatives ranked by entailment whenever this is possible (a requirement meant to derive strong positive presuppositions in cases where we observe them – unsuccessfully as we noted). By contrast, the distributed analysis of *only* faces no such explanatory burden: on that analysis, independent constraints on pruning and assertion jointly ensure that strong positive presuppositions arise by default, while weak positive presuppositions surface only in cases where the assertion would otherwise be pathological.

5 Exceptives

The preceding discussion raises several questions, including questions about expressions closely allied to *only*, such as exceptives. We show how the distributed analysis can account for some distributional differences between *only* and exceptives, as well as why our assumptions about

pruning do not interfere with the application of exhaustification in this domain – in particular, with the explanation of the idiosyncratic distribution of exceptives.

5.1 Exceptives, their distribution, and pruning

Building on von Stechow & Iatridou 2007, the distributed analysis of *only* takes *only* sentences to be exceptive constructions. This accords with the equivalence of sentences with *only* and those with negative universal quantifiers that contain exceptives, as exemplified in (132). Moreover, both *only* and exceptives exhibit identical split-scope readings in matrix and modal environments (see Sect. 1.3), and the same positive inference projection patterns. These parallels, first noted by Horn 1972, have recently been explored by Crnič 2024.

- (132) a. Only Gal_F was at the party.
 b. No one but Gal_F was at the party.
Positive inference: Gal was at the party
Exclusive inference: No one distinct from Gal was at the party

A distributed analysis of exceptives

On the distributed analysis, the sentences in (132) receive a similar treatment. Their primary difference is that the exceptive construction subtracts individuals from the nominal quantifier's domain, a meaning encoded by *but*. The structure of the exceptive sentence above is provided in (133b), assuming the complement of *but* is focused (cf. Gajewski 2013). The focus alternatives over which *exh* ranges are provided in (134a); all of the alternatives in which individuals other than Gal are subtracted from the domain of *no* are excludable, as provided in (134b).

- (133) a. No one but Gal_F was at the party.
 b. [exh_C [_S [no one [but Gal_F]] was at the party]]
- (134) a. $F(S) = \{[[\text{no one [but } X_F]] \text{ was at the party}] \mid X \in F(\text{Gal}_F)\}$
 b. $\text{IE}(S) = \{[[\text{no one [but } X_F]] \text{ was at the party}] \mid X \in F(\text{Gal}_F) \wedge [[X] \otimes \text{Gal}]\}$

The meaning of the structure in (133) is, accordingly, the one provided in (135)-(136). The presupposition in (135) is that Gal was at the party – the negation of all the excludable alternatives yields a conjunction of inferences that someone from a domain that includes Gal was at the party; among these, the inference that someone from a domain containing just Gal was at the party entails all others. The assertive meaning in (136) is that no one distinct from Gal was at the party (see, e.g., Gajewski 2013, Hirsch 2016, Crnič 2018, 2024 for discussion and derivation).⁶

⁶Pruning is not admitted in the felicitous examples of exceptive sentences. For example, the domain of *exh* in (ia) cannot be resolved as in (ii), which would yield the positive presupposition in (iii).

- (i) a. [exh_C [_S no one but Gal_F was at the party]]
 b. $C \cap \text{IE}(S) = \{[\text{no one but Tal}_F \text{ was at the party}]\}$
- (ii) Presupposition of (i):
 $\exists x: x \otimes \text{Tal} \wedge x \text{ was at the party}$

The reason is that there are other resolutions of the domain of *exh* that constitute a proper superset of the domain in (i), on which the accommodated meanings of the parses would be equivalent with that of (i), namely, that Gal was at the party and no one else – in violation of the Weakening Constraint. One such domain is provided in (iii).

- (135) Presupposition of (133b):
 $\neg \exists S' \in \text{IE}(S): \llbracket S \rrbracket = 1$
 $\Leftrightarrow \neg \exists x: x \otimes \text{Gal} \wedge \neg \exists y: y \otimes x \wedge y \text{ was at the party}$
 $\Leftrightarrow \forall x: x \otimes \text{Gal} \rightarrow \exists y: y \otimes x \wedge y \text{ was at the party}$
 $\Leftrightarrow \text{Gal was at the party}$

- (136) Assertion of (133b):
 $\neg \exists x: x \otimes \text{Gal} \wedge x \text{ was at the party}$

On the restricted distribution of exceptives

Exceptives like *but Gal* have a restricted distribution: they can combine with (negative) quantifier DPs, but not with existential quantifier DPs, as exemplified in (137). This is accounted for on the distributed analysis (esp., Gajewski 2008, 2013, building on von Stechow 1993).

- (137) #Someone but Gal_F was at the party.

The unacceptability of the exceptive in (137) is attributed to the vacuity of the accompanying *exh* operator (as argued by Gajewski 2013, Hirsch 2016, Crnič 2018, 2024, among others). More specifically, a sentence like (137) cannot be parsed so as to satisfy the condition in (138) (cf. Fox & Spector 2009, 2018, Chierchia et al. 2011; see also fn. 3 above). This same condition has been argued to underlie the distribution of a range of other polarity-sensitive expressions (see, e.g., Chierchia 2013, Ch. 3 Appendix; Spector 2014, Nicolae 2017). (The definition of *exh* in (9) precludes using it to account for restricted distributions via contradiction, as *exh* cannot generate contradictions, though see Gajewski 2009 for a discussion of some exceptional cases.)

- (138) **An economy constraint on the distribution of *exh*:**
 An occurrence of *exh* in a given sentence *S* is not licensed if deleting this occurrence leads to a sentence *S'* such that *S'* entails *S*.

The sentence in (137), with the structure in (139), violates the Economy Constraint in (138), since the sister of *exh* contains no excludable alternatives, as stated in (140). In this configuration, accordingly, exhaustification has nothing to negate.

- (139) $[\text{exh}_C \llbracket [\text{some} [\text{but Gal}_F]] \text{ was at the party} \rrbracket]$

- (140) a. $F(S) = \{ \llbracket [\text{some} [\text{but } X_F]] \text{ was at the party} \rrbracket \mid X \in F(\text{Gal}_F) \}$
 b. $\text{IE}(S) = \emptyset$

In order to demonstrate that sentence (139) indeed has no excludable alternatives, consider a maximal set of focus alternatives to the sister of *exh* that can be jointly negated with the sister of *exh* being true, that is, with someone distinct from Gal having been at the party. Suppose that someone distinct from Gal having been at the party is true because Tal was at the party. The maximal set of alternatives that can be negated in this case consists of all those that convey that someone distinct from Tal (and pluralities containing Tal) was at the party. The same holds, analogously, for all other potential verifiers of the sentence. The intersection of all these maximal sets of negatable alternatives is empty. Consequently, no focus alternative is excludable, and pruning cannot apply. This demonstrates that pruning cannot rescue violations of the Economy Constraint on

- (iii) $C \cap \text{IE}(S) = \{ \llbracket [\text{no one but } X_F \text{ was at the party}] \rrbracket \mid X \in \{ \text{Tal}, \text{Sal} \} \}$

exh, and thus cannot rescue unacceptable occurrences of exceptives. The same reasoning extends to other expressions whose distribution has been claimed to be governed by *exh*.

5.2 A puzzle and its expansion

Analyzing *only* and exceptive sentences in parallel leads to an expectation of similar variability in their positive inferences. While this parallel variability is not immediately apparent, closer scrutiny reveals that it does surface in certain cases – in ways that both support and further clarify the distributed analysis.

Alonso-Ovalle & Hirsch’s puzzle

Alonso-Ovalle & Hirsch 2023 argue that exceptives, unlike *only*, invariably trigger strong positive presuppositions. For instance, as we have seen, *only*-sentences are acceptable even when their sister is contextually incompatible with its alternatives, as repeated in (141a)–(142a). This does not extend to exceptives: when the predicate holding of the excepted element is contextually incompatible with it holding of other domain elements, the sentence becomes pragmatically marked, as shown in (141b)–(142b). This suggests that exceptives always trigger strong positive presuppositions – e.g., that Mary won the bronze in (142b) – which then result in a violation of the Informativity Constraint.

- (141) a. Gal *only* got her BA at Cal State_F.
 b. ??Gal got her BA at no place but Cal State_F.
- (142) a. Mary *only* won the bronze_F.
 b. ??Mary won no medal but the bronze_F.

Since the treatment of *only* and exceptive sentences are parallel on the distributed analysis, Alonso-Ovalle & Hirsch 2023 argue that the asymmetry in the acceptability of the above data provides grounds to reject the distributed analysis of weak positive presuppositions (see the discussion of their alternative analysis in Sect. 4.1 above).

Alonso-Ovalle & Hirsch’s observations, however, do not fully capture the empirical landscape, as there are occurrences of exceptives that behave just like the *only* examples we discussed.

Expanding the puzzle: more on exceptives

The distinction between *only* and exceptives is not as clear-cut as the discussion above might suggest. As two reviewers point out, there are occurrences of exceptives that do behave just like *only*. A similar observation was also made by Kai von Stechow (cited by Iatridou & Zeijlstra 2021, fn. 26, who also bring up parallel examples from Greek). Some examples are provided in (143).

- (143) a. Gal is nothing but a poor grad student.
 b. We are nothing but soldiers.
 c. Tal got nothing but the bronze medal.

These examples give rise not only to scalar inferences typically associated with weak positive presuppositions (see Sect. 2), but also to weak positive presuppositions. This can be demonstrated on their projective behavior in questions. For instance, the question in (144), based on (143a), does not trigger the strong positive inference that Gal is a grad student.

- (144) Is Gal nothing but a poor grad student?
 $\nRightarrow$ Gal is a poor grad student

We conclude that exceptives and *only* sentences are not entirely different after all when it comes to the availability of weak positive presuppositions.

5.3 An account of the expanded puzzle

What distinguishes Alonso-Ovalle & Hirsch's infelicitous examples of exceptives from the felicitous ones brought up by the reviewers and Kai von Fintel? The contrast is repeated below.

- (145) a. ??Gal won no medal but the bronze.
 b. Gal is nothing but a poor grad student.

One salient difference between the examples pertains to the narrowness of domain restriction of the negative quantifiers. This restriction is imposed either through overt nominals (e.g., *medal*) or through predicates that lexically constrain the domain of the nominal (e.g., *to get a BA from*). In these cases of narrow domain restriction, the strong positive presupposition (e.g., that Gal won the bronze) entails the assertive content, violating the Informativity Constraint. By contrast, felicitous examples feature maximally broad domains, typically occurring in copular constructions where the nominal complement is a general term like *thing*. We show below that the distributed analysis can explain these differences.

Containment and exceptives

On a more comprehensive distributed analysis of exceptives, exceptives have been argued to give rise to three inferences, not just the two mentioned in (132) (e.g., Hoeksema 1990, von Fintel 1993, 1994). The third inference is that the excepted element must be contained in the initial domain of quantification – the containment inference (Gajewski 2013, Vostrikova 2021, Mayr & Vostrikova 2023, e.g., argue that it is a presupposition):

- (146) No student but Gal was at the party.

Containment inference: Gal is a student

Positive inference: Gal was at the party

Exclusive inference: No student distinct from Gal was at the party

The containment inference is consequential for the variability of the strong positive inference. In particular, a weak positive inference for the marked exceptive sentence *Gal won no medal but the bronze_F* would have to be derived from negation of the alternative whose meaning corresponds to Gal having won no medal, that is, an alternative of the form *[Gal won no medal but the X_F]*, where *X* is neither *bronze*, *silver* nor *gold*. We suggest that the difficulty of accommodating additional medal types in this context prevents the derivation of such a weak positive presupposition – specifically, since the containment requirement restricts the alternatives to those built on phrases denoting actual medals, none is available. (See the Excursus at the end of Sect. 3.3 on why pruning just some of these actual alternatives is not admitted by the principle of Relevance.)

- (147) a. $[[\text{medal}]] = \{\text{the bronze, the silver, the gold}\}$
 b. $F([\text{Gal won no medal but the bronze}_F]) =$
 $\{[\text{Gal won no medal but the NP}] \mid \text{NP} \in \{\text{bronze, silver, gold}\}\}$

The situation is substantially different in the predicational example in which the complement of the negative determiner is *thing*. In this case, the salient predicates in the domain of the negative indefinite may well include *grad student*, *postdoc* and *professor*, but nothing in the structure of the sentence prevents one from accommodating additional predicates, say, *non-academic*.

- (148) a. $[[\text{thing}]] = \{\text{grad student, postdoc, professor, non-academic}\}$
 b. $F([\text{Gal is nothing but a grad student}_F]) = \{[\text{Gal is nothing but a NP}] \mid \text{NP} \in \{\text{grad student, ... non-academic, ...}\}\}$

Accordingly, an appropriate domain restriction of *exh* can result in this case in a weaker positive presupposition. For example, adopting an analogous domain restriction that we employed in *only*-sentences, provided in (149), yields the weak positive presupposition in (151), as desired.

- (149) a. $[\text{exh}_C [\text{S Gal is nothing but a grad student}_F]]$
 b. $C \cap \text{IE}(S) = \{[\text{Gal is nothing but a non-academic}]\}$

- (150) Assertion of (149a) on the resolution in (149b):
 $\neg(\text{Gal is a postdoc}) \wedge \neg(\text{Gal is a professor}) \wedge \neg(\text{Gal is a non-academic})$

- (151) Presupposition of (149a) on the resolution in (149b):
 $(\text{Gal is a grad student} \vee \text{Gal is a postdoc} \vee \text{Gal is a professor})$

A difference with *only*

Now, turning back to *only*, we imposed no containment requirement on it (cf. Bar-Lev & Fox 2020, fn. 20, for a related assumption about *exh*). Accordingly, we do not expect the sister of *only* to be in the domain of *only* in all the focus alternatives to the sister of *exh*. This allows us to derive weak positive presuppositions. While some sort of a containment condition could be operative also in the case of *only/exh* (see, e.g., Rooth 1992, Magri 2009, Fox & Katzir 2011), our proposal requires it to be restricted to the asserted material and not to extend to alternatives, as provided in (152). All our representations above satisfy this requirement. (Note that this latter requirement resembles conventional implicatures. See Crnič 2012 on non-projection of conventional implicatures from alternatives, and Sauerland 2013, Bassi 2021 for related discussion.)

- (152) ***Only, exh, and limited containment:***
 An assertion of LFs of the form $[\text{exh}_C S]$ and $[\text{only}_C S]$ is felicitous only if *S* is contained in the domain *C* of *exh* and *only*, respectively. This condition is not active in the focus alternatives to these sentences.

Summary

Although the behavior of *only* and exceptives is closely related, involving identical or very similar operations, it comes apart in relation to the nature of subtraction. In the case of exceptives, subtraction cannot be vacuous, as it is hardwired into the definition of *but*, and hence holds in all the focus alternatives to an exceptive sentence. In contrast, no such requirement applies in the focus alternatives to *only* sentences. This affects what focus alternatives to the two types sentences are licit, and consequently what positive inferences these sentences can generate.

- (153) **Parameter of variation:**

- a. The exceptive marker presupposes that its first argument is contained in its second argument. This may necessitate strong positive presuppositions.
- b. *Only* does not presuppose that its sister is in its domain (though this may be implicated about the asserted sentence). This allows for weak positive presuppositions.

6 Taking stock

Only gives rise to a weak positive presupposition when the alternatives to its sister are mutually incompatible, and a strong positive presupposition in all other cases (esp., Klinedinst 2005). We demonstrated that the distributed analysis of *only* captures this intricate behavior in a principled way, while integrated analyses have difficulty doing so. The distributed analysis achieves this because it derives the positive presupposition via exhaustification, which makes it possible to modulate its strength by pruning certain alternatives over which exhaustification quantifies. The analysis must make one concession, however: the Weakening Constraint on pruning, which prevents pruning of alternatives whose removal does not properly weaken the exhaustified meaning, can be violated when this is necessary to make the sentence assertable. The derivations of the different readings proceed as follows:

Strong positive presuppositions. In sentences in which the alternatives to the sister of *only* are mutually contextually compatible, for instance, *Only Gal_F was at the party*, strong positive presuppositions are generated. This is because the domain of *exh* must encompass all the focus alternatives to the sister of *exh*. On this resolution, the sentence satisfies the Informativity Constraint, and so the Weakening Constraint must be satisfied as well. Since any pruning of focus alternatives from the domain of *exh* violates the Weakening Constraint, no pruning is permitted, and a strong positive presupposition is generated.

- (154) **Parse:** $[exh_{C'} [only_C [Gal \text{ was at the party}]]]$
Obligatory: $F([only_C [Gal_F \text{ was at the party}]])) \subseteq C'$
 $\Rightarrow$ **Strong positive presupposition:** Gal was at the party

Weak positive presuppositions. In sentences in which the alternatives to the sister of *only* are mutually contextually incompatible, for instance, *Gal is only a grad student_F*, weak positive presuppositions are generated because the domain of *exh* may consist merely of the strongest focus alternative(s) to the sister of *exh*. Only on such a resolution does the meaning of the sentence satisfy the Informativity Constraint, which overrides the concurrent violation of the Weakening Constraint.

- (155) **Parse:** $[exh_{C'} [only_C [Gal \text{ is a } [grad \text{ student}]_F]]]$
Obligatory: $C' \cap F([only_C [Gal \text{ is a } [grad \text{ student}]_F]]) =$
 $\{([only_C [Gal \text{ is a } [grad \text{ student}]_F]], [only_C [Gal \text{ is a non-academic}_F]]\}$
 $\Rightarrow$ **Weak positive presupposition:** Gal is a grad student, a postdoc, or a professor

Exceptives. Finally, some exclusive-like expressions generate only strong positive presuppositions in certain cases, due to their imposing a containment requirement. This requirement compromises the derivation of the weak positive presupposition in some cases, as it renders the required

alternatives undefined. Notably, exceptives introduce such a containment requirement in all alternatives (as a presupposition of the exceptive marker), whereas *only* does not.

- (156) **Parse:** $\#[\text{exh}_{C'} [\text{Gal won no medal but the bronze}_F]]$
Obligatory: $F([\text{Gal won no medal but the bronze}_F]) \subseteq C'$
 $\Rightarrow$ **Strong positive presupposition, *Informativity Constraint**

Appendix: Another derivation of strong positive presuppositions

Our derivation of strong positive presuppositions in the main text focused on cases in which the associate of *only* and *exh* are proper names, and it relied on the assumption that proper names are full DPs that, accordingly, have (plural) definite descriptions as alternatives (Sect. 1.2). This assumption is neither necessary nor does it naturally generalize to cases in which the associates are predicates (such as VPs, adjectives, or nouns). In the following, we show how the alternatives required in our derivation can be obtained more generally.

Consider the simple, schematic LF in (157a), where the associate of *only* is phrase *P* in sentence *S*, which has without loss of generality just two relevant focus alternatives *Q* and *R* (the structure $S(X)$ is just like $S(P)$ except that the focused occurrence of *P* is replaced by *X*).

- (157) a. $[\text{exh}_{C'} [\text{only}_C S(P_F)]]$
b. $C \cap F(S(P_F)) = \{S(P_F), S(Q_F), S(R_F)\}$

The alternatives in the domain of *exh* are determined by the structure of its sister, esp., by what elements are focused. Accordingly, as in the main text, they will include the alternatives in (158a). More generally, however, they will also include the subdomain alternatives in (158b) – that is, alternatives in which the domain of *only* is replaced by its subdomains.

- (158) a. $\{[\text{only}_C [S(Q_F)]], [\text{only}_C [S(R_F)]]\} \subseteq \text{ALT}([\text{only}_C [S(P_F)]])$
b. $\{[\text{only}_{C*} [S(Q_F)]], [\text{only}_{C*} [S(R_F)]] \mid \llbracket C^* \rrbracket \subseteq \llbracket C \rrbracket\} \subseteq \text{ALT}([\text{only}_C [S(P_F)]])$

The meanings of the alternatives to the *only*-sentence thus include all those available prior to considering the subdomain alternatives, provided in (159a): for example, $\neg(\llbracket S(P) \rrbracket \vee \llbracket S(R) \rrbracket)$ is the meaning of $[\text{only}_C S(Q_F)]$. In addition, they now also include the meanings of subdomain alternatives: for example, $\neg\llbracket S(P) \rrbracket = \llbracket [\text{only}_{C*} [S(Q_F)]] \rrbracket = \llbracket [\text{only}_{C*} [S(R_F)]] \rrbracket$, where $\llbracket C^* \rrbracket \cap F(S(P_F)) = \{S(P)\}$.

- (159) a. $\neg(\llbracket S(P) \rrbracket \vee \llbracket S(R) \rrbracket), \neg(\llbracket S(P) \rrbracket \vee \llbracket S(Q) \rrbracket), \neg(\llbracket S(Q) \rrbracket \vee \llbracket S(R) \rrbracket)$
b. $\neg\llbracket S(P) \rrbracket, \neg\llbracket S(Q) \rrbracket, \neg\llbracket S(R) \rrbracket$

The excludable alternatives to $[\text{only}_C [S(P_F)]]$ have the meanings provided in (160). The meaning of the sentence, on the assumption that no pruning applies, is provided in (160) – the presupposition is equivalent to the meaning of $S(P)$, the strong positive presupposition, as desired.

- (160) $\neg(\llbracket S(P) \rrbracket \vee \llbracket S(R) \rrbracket), \neg(\llbracket S(P) \rrbracket \vee \llbracket S(Q) \rrbracket), \neg\llbracket S(P) \rrbracket$

- (161) a. Assertion of (157a): $\neg\llbracket S(Q) \rrbracket \wedge \neg\llbracket S(R) \rrbracket$
b. Presupposition of (157a): $(\llbracket S(P) \rrbracket \vee \llbracket S(R) \rrbracket) \wedge (\llbracket S(P) \rrbracket \vee \llbracket S(Q) \rrbracket) \wedge \llbracket S(P) \rrbracket$
 $\Leftrightarrow \llbracket S(P) \rrbracket$

References

- Abusch, Dorit. 2010. Presupposition triggering from alternatives. *Journal of Semantics* 27(1). 37–80.
- Alonso-Ovalle, Luis. 2008. Innocent exclusion in an alternative semantics. *Natural Language Semantics* 16(2). 115–128. doi:10.1007/s11050-008-9027-1. <https://doi.org/10.1007/s11050-008-9027-1>.
- Alonso-Ovalle, Luis & Aron Hirsch. 2022. Keep only strong. *Semantics and Pragmatics* 15. 6–1.
- Alonso-Ovalle, Luis & Aron Hirsch. 2023. Weakening is external to only. In *Semantics and linguistic theory*, 583–602.
- Alxatib, Sam. 2020. *Focus, evaluativity, and antonymy*. Springer.
- Anvari, Amir. 2018. Logical integrity. In *Semantics and linguistic theory*, vol. 28, 711–726.
- Atlas, Jay David. 1993. The importance of being ‘only’: Testing the neo-gricean versus neo-entailment paradigms. *Journal of Semantics* 10(4). 301–318.
- Bade, Nadine & Konstantin Sachs. 2019. Exh passes on alternatives: A comment on fox and spectator (2018). *Natural Language Semantics* 27(1). 19–45.
- Bar-Lev, Moshe E. 2024. Obligatory implicatures and the relevance of contradictions. *Journal of Semantics* 41(2). 149–173.
- Bar-Lev, Moshe E & Danny Fox. 2020. Free choice, simplification, and innocent inclusion. *Natural Language Semantics* 28(3). 175–223.
- Bassi, Itai. 2021. *Fake feature and valuation from context*: Massachusetts Institute of Technology dissertation.
- Bassi, Itai, Guillermo Del Pinal & Uli Sauerland. 2021. Presuppositional exhaustification. *Semantics and Pragmatics* 14. 11. <https://semprag.org/article/view/sp.14.11>.
- Beaver, David & Brady Z. Clark. 2008. *Sense and sensitivity: How focus determines meaning*. Blackwell.
- Beaver, David & Emiel Krahmer. 2001. A partial account of presupposition projection. *Journal of Logic, Language and Information* 10(2). 147.
- Beck, Sigrid. 2009. Multiple focus. *Journal of Semantics* 26. 159–184.
- Beck, Sigrid. 2016. Discourse related readings of scalar particles. In *Semantics and linguistic theory*, vol. 26, 142–165.
- Borer, Hagit. 2005. *Structuring sense, volume i: In name only*. Oxford: Oxford University Press.
- Buccola, Brian & Andreas Haida. 2019. Obligatory irrelevance and the computation of ignorance inferences. *Journal of Semantics* 36(4). 583–616.
- Chierchia, Gennaro. 2013. *Logic in grammar*. Oxford: Oxford University Press.

- Chierchia, Gennaro, Danny Fox & Benjamin Spector. 2011. The grammatical view of scalar implicatures and the relationship between semantics and pragmatics. In Paul Portner, Claudia Maienborn & Klaus von Stechow (eds.), *Handbook of semantics*, Mouton de Gruyter.
- Cohen, Ariel & Manfred Krifka. 2014. Superlative quantifiers and meta-speech acts. *Linguistics and philosophy* 37(1). 41–90.
- Condoravdi, Cleo & Itamar Francez. 2022. A minimally sufficient analysis of sufficiency modal constructions. In *Proceedings of the amsterdam colloquium*, 84–90.
- Coppock, Elizabeth & David Beaver. 2014. Principles of the exclusive muddle. *Journal of Semantics* 31(3). 371–432.
- Crnič, Luka. 2011. On the meaning and distribution of concessive scalar particles. In N. LaCara, L. Fainleib & Y. Park (eds.), *Proceedings of NELS 41*, 143–156.
- Crnič, Luka. 2012. Projection of supplements in alternatives. *Snippets* 26. <https://www.ledonline.it/snippets/allegati/snippets26001.pdf>.
- Crnič, Luka. 2013. Focus particles and embedded exhaustification. *Journal of semantics* 30(4). 533–558.
- Crnič, Luka. 2018. A note on connected exceptives and approximatives. *Journal of Semantics* 35(4). 741–756. <https://doi.org/10.1093/jos/ffy011>.
- Crnič, Luka. 2022. A distributed analysis of *only* [first, longer version]. Manuscript, HUJI. <https://ling.auf.net/lingbuzz/006645>.
- Crnič, Luka. 2024. A distributed analysis of *only*. *Linguistics and Philosophy* 47(4). 611–652.
- Crnič, Luka, Emmanuel Chemla & Danny Fox. 2015. Scalar implicatures of embedded disjunction. *Natural Language Semantics* 23(4). 271–305.
- Del Pinal, Guillermo, Itai Bassi & Uli Sauerland. 2024. Free choice and presuppositional exhaustification. *Semantics and Pragmatics* 17. 3–1.
- Doron, Omri & Jad Wehbe. 2022. A constraint on presupposition accommodation. In *Proceedings of the 23rd amsterdam colloquium*, vol. 23, .
- Elbourne, Paul. 2013. *Definite descriptions*, vol. 1. Oxford University Press, USA.
- Elbourne, Paul D. 2005. *Situations and individuals*, vol. 90. MIT press Cambridge, MA.
- Enguehard, Émile. 2021. Minimal sufficiency readings of necessity modals. Manuscript, ENS. <https://ling.auf.net/lingbuzz/006826>.
- Fălăuş, Anamaria. 2014. (partially) free choice of alternatives. *Linguistics and Philosophy* 37(2). 121–173. doi:10.1007/s10988-014-9148-1.
- von Stechow, Kai. 1993. Exceptive constructions. *Natural Language Semantics* 1(2). 123–148.
- von Stechow, Kai. 1994. *Restrictions on quantifier domains*: University of Massachusetts dissertation.

- von Fintel, Kai. 1997. Bare plurals, bare conditionals, and only. *Journal of Semantics* 14(1). 1–56.
- von Fintel, Kai. 2001. Counterfactuals in a dynamic context. In Michael Kenstowicz (ed.), *Ken Hale: a life in language*, 123–152. MIT Press.
- von Fintel, Kai. 2024. Enough! In Eva Csipak Lisa Matthewson Marcin Morzycki Daniel Reisinger Bochnak, Ryan (ed.), *The title of this volume is shorter than its contributions are allowed to be: Papers in honour of hotze rullmann*, 145–156.
- von Fintel, Kai & Sabine Iatridou. 2005. Anatomy of a modal. In Bernhard Nickel Seth Yalcin Jon Gajewski, Valentine Hacquard (ed.), *New work on modality*, vol. 51, MITWPL.
- von Fintel, Kai & Sabine Iatridou. 2007. Anatomy of a modal construction. *Linguistic Inquiry* 38(3). 445–483.
- Fox, Danny. 1999. Focus, parallelism and accommodation. In *Semantics and linguistic theory*, vol. 9, 70–90.
- Fox, Danny. 2007. Free choice and the theory of scalar implicatures. In Uli Sauerland & Penka Stateva (eds.), *Presupposition and Implicature in Compositional Semantics*, 71–120. Palgrave Macmillan.
- Fox, Danny. 2016. On why ignorance might be part of literal meaning: Commentary on Marie-Christine Meyer. Handout of a talk delivered at MIT Workshop on Exhaustivity, Cambridge, MA. <https://sites.google.com/site/exhaustivity2016/>.
- Fox, Danny. 2020. Pointwise exhaustification and the semantics of question embedding. Manuscript, MIT. <https://semanticsarchive.net/Archive/jc5NmIxN/Question%20Embedding>.
- Fox, Danny & Martin Hackl. 2006. The universal density of measurement. *Linguistics and Philosophy* 29. 537–586.
- Fox, Danny & Roni Katzir. 2011. On the characterization of alternatives. *Natural Language Semantics* 19(1). 87–107.
- Fox, Danny & Benjamin Spector. 2009. Economy and embedded exhaustification. Handout from a talk at Cornell. MIT & ENS.
- Fox, Danny & Benjamin Spector. 2018. Economy and embedded exhaustification. *Natural language semantics* 26(1). 1–50.
- Gajewski, Jon. 2008. NPI any and connected exceptive phrases. *Natural Language Semantics* 16(1). 69–110.
- Gajewski, Jon. 2009. Innocent exclusion is not contradiction free. *Manuscript, University of Connecticut*.
- Gajewski, Jon. 2013. An analogy between a connected exceptive phrase and polarity items. In *Beyond any and ever*, vol. 262, 183–212. Walter de Gruyter.
- Geurts, B. & R. van der Sandt. 2004. Interpreting focus. *Theoretical Linguistics* 30. 1–44.

- Geurts, Bart. 1997. Good news about the description theory of names. *Journal of semantics* 14(4). 319–348.
- Gillies, Anthony. 2007. Counterfactual scorekeeping. *Linguistics and Philosophy* 30(3). 329–360.
- Greenberg, Yael. 2022. On the scalar antonymy of only and even. *Natural Language Semantics* 30(4). 415–452.
- Grice, Paul. 1975. Logic and Conversation. In Peter Cole & J.L. Morgan (eds.), *Syntax and semantics*, vol. 3, 41–58. Academic Press.
- Groenendijk, Jeroen Antonius Gerardus & Martin Johan Bastiaan Stokhof. 1984. *Studies on the semantics of questions and the pragmatics of answers*: Univ. Amsterdam dissertation.
- Grosz, Patrick Georg. 2012. *On the grammar of optative constructions*, vol. 193. John Benjamins Publishing.
- Guerrini, Janek & Jad Wehbe. 2025. Homogeneity as presuppositional exhaustification. *Journal of Semantics* 42(3). 281–301. doi:10.1093/jos/ffaf007. <https://doi.org/10.1093/jos/ffaf007>.
- Hirsch, Aron. 2016. An unexceptional semantics for expressions of exception. In *University of pennsylvania working papers in linguistics*, vol. 22 1, <https://repository.upenn.edu/pwpl/vol22/iss1/16/>.
- Hirsch, Aron. 2017. *An inflexible semantics for cross-categorial operators*: Massachusetts Institute of Technology dissertation.
- Hoeksema, Jack. 1990. Exploring exception phrases. In *Proceedings of the seventh amsterdam colloquium*, 165–90. ITLI Amsterdam.
- Horn, Larry R. 1969. A presuppositional theory of ‘only’ and ‘even’. In *Proceedings of Chicago Linguistics Society* 5, .
- Horn, Larry R. 1972. *On the semantic properties of logical operators in English*: UCLA dissertation.
- Horn, Larry R. 1992. The said and the unsaid. In Chris Barker & David Dowty (eds.), *Proceedings of SALT 2*, .
- Horn, Larry R. 1996. Exclusive company: Only and the dynamics of vertical inference. *Journal of Semantics* 13(1). 1–40. doi:10.1093/jos/13.1.1.
- Iatridou, Sabine & Hedde Zeijlstra. 2021. The complex beauty of boundary adverbials: In years and until. *Linguistic Inquiry* 52(1). 89–142. doi:10.1162/ling.2021.52.1.89. <https://direct.mit.edu/ling/article/52/1/89/106746>.
- Ippolito, Michela. 2007. On the meaning of only. *Journal of Semantics* 25(1). 45–91.
- Ippolito, Michela. 2008. On the meaning of only. *Journal of Semantics* 25(1). 45–91.
- Jackendoff, R.S. 1972. *Semantic interpretation in generative grammar*. The MIT Press.
- Jacobs, Joachim. 1983. *Fokus und Skalen*. Niemayer.

- Katzir, Roni. 2007. Structurally defined alternatives. *Linguistics and Philosophy* 30. 669–690.
- Katzir, Roni. 2014. On the roles of markedness and contradiction in the use of alternatives. In Salvatore Pistoia-Reda (ed.), *Pragmatics, semantics and the case of scalar implicatures*, 40–71. Springer.
- Klinedinst, Nathan. 2005. Scales and ‘only’. Unpublished ms. UCLA.
- Krasikova, Sveta & Ventsislav Zhechev. 2006. You only need a scalar ‘only’. *ZAS Papers in Linguistics* 44(1). 199–209.
- Kratzer, Angelika. 1986. Conditionals. In Arnim von Stechow & Dieter Wunderlich (eds.), *Semantics: An international handbook of contemporary research*, 651–656. Mouton de Gruyter.
- Krifka, Manfred. 1993. Focus and presupposition in dynamic interpretation. *Journal of semantics* 10(4). 269–300.
- Krifka, Manfred. 2014. Embedding illocutionary acts. In *Recursion: Complexity in cognition*, 59–87. Springer.
- Lewis, D. 1973. *Counterfactuals*. Wiley-Blackwell.
- Lewis, David. 1988. Relevant implication. *Theoria* 54(3). 161–174.
- Longobardi, Giuseppe. 1994. Reference and proper names. *Linguistic Inquiry* 25(4). 609–665.
- Magri, Giorgio. 2009. A theory of individual level predicates based on blind mandatory scalar implicatures. *Natural Language Semantics* 17. 245–297.
- Magri, Giorgio. 2011. Another argument for embedded scalar implicatures based on oddness in downward entailing environments. *Semantics and Pragmatics* 4(6). 1–51.
- Mayr, Clemens. 2014. Intervention effects and additivity. *Journal of Semantics* 31(4). 513–554. doi:10.1093/jos/fft010.
- Mayr, Clemens & Ekaterina Vostrikova. 2023. A unified semantics for exceptive-additive ‘besides’. In *Proceedings of sinn und bedeutung*, vol. 27, 399–416.
- McCawley, James D. 1993. *Everything that linguists have always wanted to know about logic... but were ashamed to ask*. University of Chicago Press.
- Meyer, Marie-Christine. 2013. *Ignorance and grammar*: MIT dissertation.
- Nicolae, Andreea C. 2017. Deriving the positive polarity behavior of plain disjunction. *Semantics & Pragmatics* 10(5). 1–21. doi:10.3765/sp.10.5. <https://semprag.org/index.php/sp/article/view/sp.10.5>.
- Panizza, Daniele & Yasutada Sudo. 2020. Minimal sufficiency with covert ‘even’. *Glossa* 5(1).
- Roberts, Craige. 2006. Only, presupposition and implicature. Ms., The Ohio State University.
- Roberts, Craige. 2011. Only: A case study in projective meaning. *The Baltic International Yearbook of Cognition, Logic and Communication* 6. 1–59.

- Rooth, Mats. 1985. *Association with focus*: University of Massachusetts, Amherst, PhD dissertation.
- Rooth, Mats. 1992. A theory of focus interpretation. *Natural Language Semantics* 1(1). 75–116.
- van Rooy, Robert & Katrin Schulz. 2007. *Only*: Meaning and Implicatures. In Maria Aloni, Alistair Butler & Paul Dekker (eds.), *Questions and Answers*, Elsevier.
- Sauerland, Uli. 2013. Presuppositions and the alternative tier. In *Semantics and linguistic theory*, 156–173.
- Schwarz, B. 2005. Scalar Additive Particles in Negative Contexts. *Natural Language Semantics* 13. 125–168.
- Singh, Raj, Ken Wexler, Andrea Astle-Rahim, Deepthi Kamawar & Danny Fox. 2016. Children interpret disjunction as conjunction: Consequences for theories of implicature and child development. *Natural Language Semantics* 24(4). 305–352. doi:10.1007/s11050-016-9126-3. <https://doi.org/10.1007/s11050-016-9126-3>.
- Spector, Benjamin. 2014. Scalar implicatures, blindness and common knowledge: Comments on magri (2011). *Semantics, Pragmatics, and the Case of Scalar Implicatures* 146–169.
- Spector, Benjamin. 2016. Comparing exhaustivity operators. *Semantics and Pragmatics* 9(11). 1–33. doi:10.3765/sp.9.11. <https://doi.org/10.3765/sp.9.11>.
- Stalnaker, Robert. 1968. A theory of conditionals. In Nicholas Rescher (ed.), *Studies in logical theory* (supplementary monograph to the American Philosophical Quarterly), 315–332. Blackwell.
- Stalnaker, Robert C. 1974. Pragmatic presuppositions. In Milton K. Munitz & Peter Unger (eds.), *Semantics and philosophy*, 197–213. New York: New York University Press.
- Stalnaker, Robert C. 1978. Assertion. In *Pragmatics*, 315–332. Brill.
- Taglicht, Josef. 1984. *Message and emphasis: On focus and scope in english* 15. Addison-Wesley Longman Limited.
- Tonhauser, Judith, David Beaver, Craige Roberts & Mandy Simons. 2013. Toward a taxonomy of projective content. *Language* 89(1). 66–109.
- Trinh, Tue. 2019. Exhaustification and contextual restriction. *Frontiers in Communication* 4. 47.
- Trinh, Tue & Andreas Haida. 2015. Constraining the derivation of alternatives. *Natural Language Semantics* 23(4). 249–270.
- Vostrikova, Ekaterina. 2021. Conditional analysis of clausal exceptives. *Natural Language Semantics* 29(2). 159–227.
- Wagner, Michael. 2006. Association by movement: evidence from np_i-licensing. *Natural Language Semantics* 14(4). 297–324.
- Zeevat, Henk. 2013. Expressing surprise by particles. In *Beyond expressives: Explorations in use-conditional meaning*, 297–320. Brill.